



14. Children and Adolescents: Standards of Care in Diabetes—2026

American Diabetes Association
Professional Practice Committee for
Diabetes*

Diabetes Care 2026;49(Suppl. 1):S297–S320 | <https://doi.org/10.2337/dc26-S014>

The American Diabetes Association (ADA) “Standards of Care in Diabetes” includes the ADA’s current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA’s clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

The management of diabetes in children and adolescents cannot simply be derived from care routinely provided to adults with diabetes. The epidemiology, pathophysiology, developmental considerations, and response to therapy in pediatric diabetes are often different from those of adult diabetes. There are also differences in recommended care for children and adolescents with type 1 diabetes, type 2 diabetes, and other forms of diabetes.

This section is organized into four major parts: the first part introduces type 1 and type 2 diabetes in children and adolescents. The second part addresses care for children and adolescents with all types of diabetes, including diabetes self-management education and support (DSMES) and nutrition therapy; the third part addresses care specifically for children and adolescents with type 1 diabetes, and the fourth part addresses care specifically for children and adolescents with type 2 diabetes. Monogenic diabetes (neonatal diabetes and maturity-onset diabetes of the young) and cystic fibrosis–related diabetes, which are often present in children and adolescents, are discussed in section 2, “Diagnosis and Classification of Diabetes.” **Table 14.1** provides recommendations for screening and treatment of complications and related conditions in pediatric type 1 diabetes and type 2 diabetes. In addition to comprehensive diabetes care, children and adolescents with diabetes should receive age-appropriate and developmentally appropriate pediatric care, including immunizations as recommended by the American Academy of Pediatrics (AAP) (1). To ensure continued care as a person with diabetes becomes an adult, guidance is provided at the end of this section on the transition from pediatric to adult diabetes care.

Expert opinion and a review of available and relevant experimental data are summarized in the American Diabetes Association (ADA) position statements “Type 1 Diabetes in Children and Adolescents” (2) and “Evaluation and Management of Youth-Onset Type 2 Diabetes” (3). Finally, other sections in the Standards of Care

*A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at <https://doi.org/10.2337/dc26-SINT>.

This section has received endorsement from The International Society for Pediatric and Adolescent Diabetes.

Duality of interest information for each contributor is available at <https://doi.org/10.2337/dc26-SDIS>.

Suggested citation: American Diabetes Association Professional Practice Committee for Diabetes. 14. Children and adolescents: Standards of Care in Diabetes—2026. *Diabetes Care* 2026; 49(Suppl. 1): S297–S320

© 2025 by the American Diabetes Association. Readers may use this work for educational, noncommercial purposes if properly cited and unaltered. The version of record may be linked at <https://diabetesjournals.org/care>, but ADA permission is required to post this work on any third-party site or platform. This publication and its contents may not be reproduced, distributed, or used for text or data mining, machine learning, or similar technologies without prior written permission. Requests to reuse, adapt, or distribute this work may be sent to permissions@diabetes.org. More information is available at <https://diabetesjournals.org/journals/pages/license>.

may have recommendations that apply to children and adolescents with diabetes and are referenced in the narrative of this section.

INTRODUCTION TO TYPE 1 DIABETES IN CHILDREN AND ADOLESCENTS

Type 1 diabetes is the most common form of diabetes in children and adolescents (4), although there are more adults living with and diagnosed with type 1 diabetes (5). Nearly 4 in every 1,000 children and adolescents in the U.S. were reported to have type 1 diabetes from 2019 through 2022 (6). The health care professional must consider the unique aspects of care and management of children and adolescents with type 1 diabetes, such as changes in insulin sensitivity related to physical growth and sexual maturation, ability to provide self-care, supervision in the childcare and school and athletic environments, emerging independence in adolescents, neurological vulnerability to hypoglycemia and hyperglycemia in young children, and adverse neurocognitive effects of diabetic ketoacidosis (DKA) (7). Attention to family dynamics, developmental stages, and physiologic differences related to sexual maturity is essential in developing and implementing an optimal diabetes treatment plan (8). Additionally, more people (adults, children, and adolescents) with type 1 diabetes are living with obesity than in the past, which adds to the complexity of living with and managing type 1 diabetes (9). Information addressing issues specific to children and adolescents with type 1 diabetes (e.g., glycemic goals, DKA, pumps and automated insulin dosing, autoimmunity) is provided in TYPE 1 DIABETES IN CHILDREN AND ADOLESCENTS, below.

INTRODUCTION TO TYPE 2 DIABETES IN CHILDREN AND ADOLESCENTS

The prevalence of type 2 diabetes in children and adolescents has continued to increase over the past 20 years (4). The U.S. Centers for Disease Control and Prevention published projections for type 2 diabetes prevalence using the SEARCH for Diabetes in Youth (SEARCH) study database. Assuming a 2.3% annual increase, the prevalence among those under 20 years of age will quadruple in 40 years (10,11). Information addressing issues specific to children and adolescents with type 2 diabetes is

provided in TYPE 2 DIABETES IN CHILDREN AND ADOLESCENTS, below.

Evidence suggests that type 2 diabetes in children and adolescents is different not only from type 1 diabetes but also from type 2 diabetes in adults and has unique features, such as a more rapidly progressive decline in β -cell function and accelerated development of diabetes complications (3,12). Long-term follow-up data from the Treatment Options for Type 2 Diabetes in Adolescents and Youth (TODAY) study reported most individuals with type 2 diabetes diagnosed as children or adolescents had microvascular complications by young adulthood (13). Type 2 diabetes disproportionately impacts youth from historically marginalized communities and can occur in complex psychosocial and cultural contexts which may make it difficult to implement and sustain healthy lifestyle changes and self-management behaviors (9,14,15). Additional risk factors include obesity and excess adiposity (16), family history of diabetes possibly mediated by shared genetics, lifestyle, and environmental factors (17), female sex, maternal gestational diabetes mellitus (18), and adverse social determinants of health (12). Information addressing issues specific to children and adolescents with type 2 diabetes (e.g., glycemic goals) is provided in TYPE 2 DIABETES IN CHILDREN AND ADOLESCENTS, below.

GUIDELINES FOR ALL CHILDREN AND ADOLESCENTS WITH DIABETES

Diabetes Self-Management Education and Support

An interprofessional team trained in pediatric diabetes management and sensitive to the challenges of diabetes management in children and adolescents, as well as their family members, should provide diabetes-specific care for this population. It is essential that DSMES, medical nutrition therapy, and psychosocial and behavioral support be provided at diagnosis and routinely (e.g., at each follow-up visit) thereafter in a developmentally appropriate format that builds on prior knowledge by a team of health care professionals experienced with the biological, educational, nutritional, behavioral, and emotional needs of the growing child and adolescent along with family members. Children and adolescents with diabetes spend much of the day in school. Therefore, close

communication with and the cooperation of school personnel are essential for optimal diabetes management and safety and maximal academic and athletic opportunities. The diabetes team should ask about and discuss diabetes management responsibilities with children, adolescents, and parents or caregivers on an ongoing basis, taking into account developmental and psychosocial needs.

Recommendation

14.1 Children and adolescents with diabetes, and their parents or caregivers (for individuals aged <18 years), should receive comprehensive, culturally sensitive, and developmentally appropriate individualized diabetes self-management education and support according to reference standards at diagnosis and routinely thereafter. **B**

Pediatric DSMES, whether for type 1 or type 2 diabetes, involves both children and adolescents and their parents or caregivers. Family involvement is a vital component of optimal diabetes management throughout childhood and adolescence. However, no matter how sound the medical plan is, self-management will only be effective if the affected individuals and/or family can implement it. In a 2024 meta-analysis of 42 randomized controlled trials (RCTs) conducted across the globe, in children and adolescents with type 1 diabetes, both educational DSMES and psychological support were found to improve A1C (19). While there are no DSMES meta-analyses on children or adolescents with type 2 diabetes, a 2024 systematic review and meta-analysis of 55 family-based type 2 diabetes prevention studies reported that lifestyle behaviors, body weight, and body fat percentage all improved following family participation in prevention-focused programs (20).

Diabetes care requires an approach that places the child or adolescent with diabetes and their parents or caregivers at the center of the care model. The care team must be capable of evaluating the educational, behavioral, emotional, and psychosocial factors that affect treatment plan implementation and must work with the children, adolescents, and family members to overcome barriers or redefine goals as appropriate. As the child with diabetes enters the

adolescent years, part of the clinical visit should include time for the health care professional to meet alone with the adolescent to begin the process of increasing autonomy.

As a child or adolescent with diabetes grows, develops, and acquires the need and desire for greater independent self-care skills, DSMES requires periodic and routine (e.g., at each follow-up visit) reassessment. The pediatric diabetes care team should work with children and adolescents with diabetes and their parents or caregivers to ensure there is not a premature transfer of self-management tasks to children and adolescents during this time. In addition, it is important to assess the educational needs and skills of, and provide training to, daycare workers, school nurses, and school personnel who are responsible for the care and supervision of the child with diabetes (21). An interprofessional diabetes team, including a physician or advanced practice health care professional (e.g., nurse practitioner, physician assistant), certified diabetes care and education specialist (CDCES), registered dietitian nutritionist, and behavioral health specialist or social worker, is essential. For children and adolescents with type 2 diabetes, initial treatment must include management of comorbidities such as obesity, dyslipidemia, hypertension, and microvascular complications along with assessment of social needs.

For children and adolescents with all types of diabetes, providing education on healthy lifestyle behaviors is critically important, and typically this is done through DSMES. For those with type 1 diabetes, insulin therapy is paramount, alongside DSMES. For treatment of type 2 diabetes in children and adolescents, lifestyle management, also taught through DSMES, is foundational, and pharmacologic treatment with or without insulin is necessary. For children and adolescents with obesity and diabetes, it should be noted that diabetes type is often uncertain in the first few weeks of treatment due to overlap in presentation and that a substantial percentage will present with clinically significant ketoacidosis (22). Therefore, initial therapy should address the hyperglycemia and associated metabolic derangements irrespective of ultimate diabetes type, with adjustment of therapy once metabolic compensation has been established and subsequent information, such as islet

autoantibody results, becomes available. **Figure 14.1** provides an approach to the initial treatment of new-onset diabetes in children and adolescents with overweight or obesity with clinical suspicion of type 2 diabetes.

Nutrition Therapy

Recommendations

14.2 Provide individualized medical nutrition therapy for children and adolescents with prediabetes or diabetes, emphasizing key nutrition principles (i.e., more nonstarchy vegetables, whole fruits, legumes, whole grains, nuts and seeds, and low-fat dairy products and less sugar-sweetened beverages, sweets, meat, refined grains, and processed or ultraprocessed foods), **B** as an essential component of the overall treatment plan.

14.3 Monitor carbohydrate intake, whether by carbohydrate counting or experience-based estimation, as a key component to optimizing glycemic and weight management for children and adolescents with diabetes. **B**

14.4 Educate children and adolescents and their caregivers on the potential need to adjust insulin with high-fat and high-protein meals. Meal composition affects postprandial glucose excursions. **A**

14.5 In children and adolescents with diabetes, provide comprehensive nutrition education at diagnosis, and at least annually and ideally more frequently, by an experienced registered dietitian nutritionist to assess the eating pattern in relation to weight status, age-appropriate growth, risk for disordered eating behaviors, and cardiovascular disease risk factors. **E**

Nutrition management for all types of diabetes in children and adolescents should be individualized and needs to consider family habits, food preferences, religious or cultural needs, finances, schedules, physical activity, and the individual's and family's abilities in numeracy, literacy, and self-management strategies. Routine visits with a registered dietitian nutritionist, preferably experienced in working with pediatric populations with diabetes, are recommended and should include assessment for changes in food preferences over time, access to food, growth and development, weight status,

cardiovascular risk, and potential for disordered eating.

Consuming an evidence-based eating pattern is associated with better glycemic outcomes in children and adolescents with type 1 and type 2 diabetes (23–25), although the evidence in type 2 diabetes is less robust. Education on healthy, culturally aligned eating should be provided for all children and adolescents with diabetes and their families. For children and adolescents taking mealtime insulin, carbohydrate content is the primary variable for calculation of mealtime insulin doses. Meals with higher fat and protein content, however, can cause early hypoglycemia and delayed postprandial hyperglycemia. Some adjustments in insulin dosing, including a split dose or an increase in calculated dose or use of an extended bolus feature when available to account for proteins and fats, can improve postprandial glucose management (26,27). Additionally, considering that carbohydrate counting in pregnant adults with type 1 diabetes has been reported to aid in weight management, as it contributes to portion management, so it is reasonable to consider it as a useful tool for both glycemic and weight management in children and adolescents (28).

Given the complex social and environmental context surrounding children and adolescents with prediabetes and diabetes, individual-level lifestyle interventions may not be sufficient to address the multifaceted interplay of family dynamics, behavioral health, community readiness, and the broader environmental system (3). In these cases, referrals to members of the care team with expertise in behavioral health and social work might be beneficial where available.

Physical Activity and Exercise

Recommendations

14.6 Physical activity is recommended for all children and adolescents with diabetes with the goal of 60 min of moderate- to vigorous-intensity aerobic activity daily, with vigorous muscle-strengthening and bone-strengthening activities at least 3 days per week. **B**

14.7 Children and adolescents with diabetes using insulin therapy and their parents should monitor glucose levels before, during, and after physical activity and receive education about personalized glycemic goals

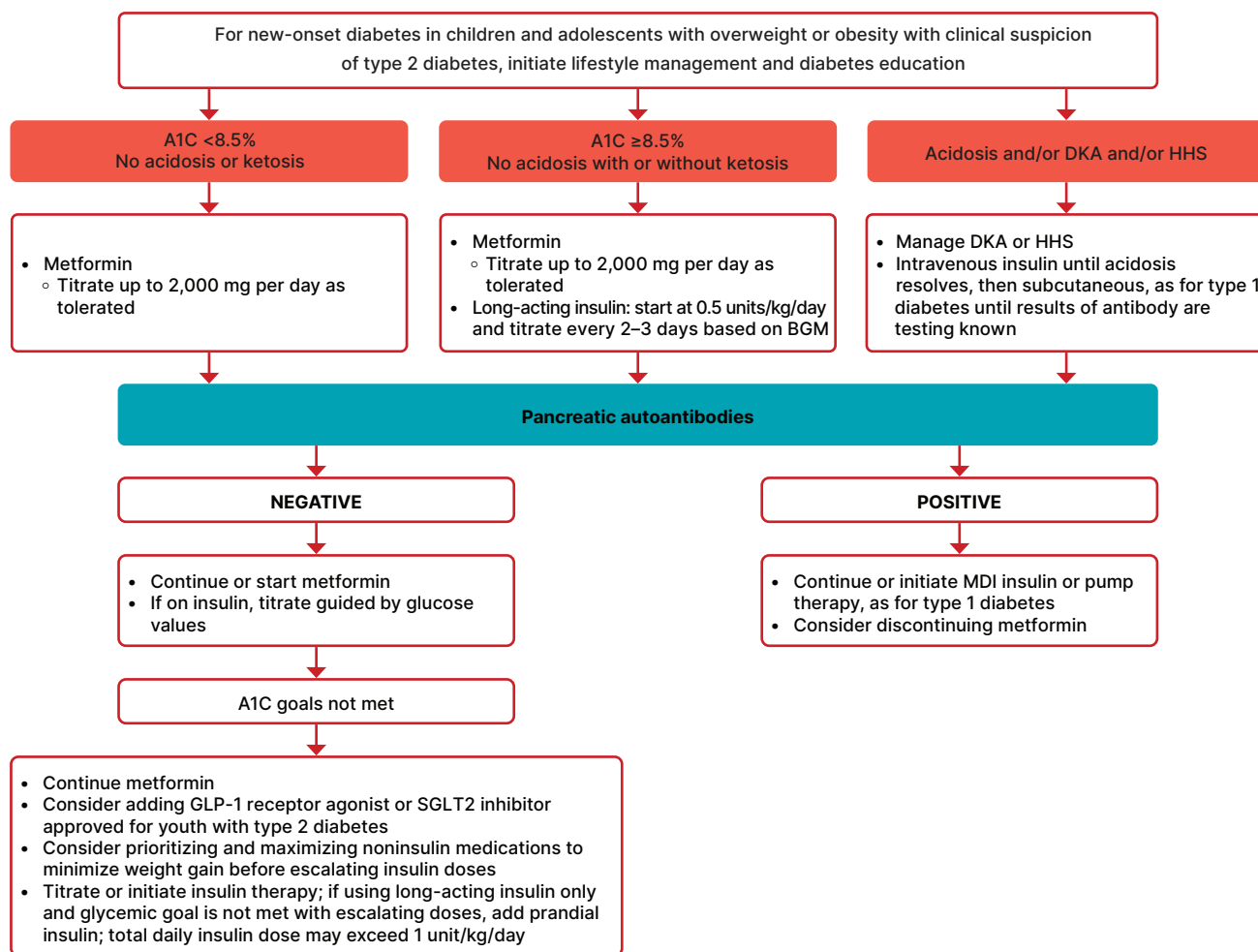


Figure 14.1—Management of new-onset diabetes in youth with overweight or obesity with clinical suspicion of type 2 diabetes. A1C 8.5% = 69 mmol/mol. BGM, blood glucose monitoring; CGM, continuous glucose monitoring; DKA, diabetic ketoacidosis; GLP-1, glucagon-like peptide 1; HHS, hyperosmolar hyperglycemic state; MDI, multiple daily injections; SGLT2, sodium–glucose cotransporter 2. Adapted from the ADA position statement “Evaluation and Management of Youth-Onset Type 2 Diabetes” (3).

according to the type and intensity of the planned physical activity. **C**
14.8 Children and adolescents with diabetes and their parents or caregivers should be educated on strategies to prevent hypoglycemia during, after, and overnight following physical activity or exercise. Treatment for hypoglycemia should be accessible before, during, and after engaging in activity. **C**

Like for all children and adolescents, physical activity and structured exercise positively impact metabolic and psychological health in children with type 1 diabetes (23,29–31), prediabetes, and type 2 diabetes (25,32–34). Exercise can have positive effects on insulin sensitivity, strength, cardiorespiratory fitness, weight management, social interaction, mood, self-esteem building, and perhaps most

importantly, the creation of healthful habits for adulthood. However, for children and adolescents using insulin, exercise also has the potential to cause both hypoglycemia and hyperglycemia, so providing education on those effects is warranted.

See below for strategies to mitigate hypoglycemia risk and minimize hyperglycemia associated with exercise in children and adolescents with type 1 diabetes or those using insulin. For an in-depth discussion, see previously published reviews and guidelines (35–37).

Overall, it is recommended that all children and adolescents, including those with prediabetes and diabetes, should participate in 60 min of moderate-intensity (e.g., brisk walking, dancing) to vigorous-intensity (e.g., running, jumping rope, soccer) aerobic activity daily, including resistance and flexibility training (34). Although

uncommon in the pediatric population, they should be evaluated for comorbid conditions or diabetes complications that may restrict participation in an exercise program.

School and Child Care

As the majority of waking hours for most children and adolescents can be spent in school and/or daycare, training of school or daycare personnel as well as athletic coaches to provide care in accordance with the individualized diabetes medical management plan is essential for optimal diabetes management and safe access to all school- or daycare-sponsored opportunities (38–40). This is of particular importance for those children and adolescents who require insulin. In addition, federal and state laws require schools, daycare facilities, and other entities to provide needed diabetes care to enable the child

to safely access the school or daycare environment. Refer to the ADA position statements “Diabetes Care in the School Setting” (38) and “Care of Young Children with Diabetes in the Childcare and Community Setting” (40) and the ADA’s Safe at School website (diabetes.org/resources/known-your-rights/safe-at-school-state-laws) for additional details.

Psychosocial Care

Recommendations

14.9 At diagnosis and during routine follow-up care, screen children and adolescents with type 1 diabetes, **B** type 2 diabetes, **B** and other forms of diabetes **E** for psychosocial concerns (e.g., diabetes distress, depressive symptoms, anxiety, disordered eating behaviors), family factors, and behavioral health concerns that could affect diabetes management with age-appropriate standardized and validated tools. Refer to a qualified behavioral health professional, preferably experienced in childhood diabetes, when indicated.

14.10 Behavioral health professionals should be considered integral members of the pediatric diabetes interprofessional team. **E**

14.11 Encourage developmentally appropriate family involvement in diabetes management tasks for children and adolescents with type 1 diabetes, type 2 diabetes, **B** and other forms of diabetes, **E** recognizing that premature or unsupportive transfer of diabetes care responsibility to children and adolescents can contribute to diabetes distress, lower engagement in diabetes self-management behaviors, and deterioration in glycemia.

14.12 Health care professionals should screen for food insecurity, housing stability, health literacy, financial barriers, and social or community support and apply that information to treatment decisions. **E**

14.13 Health care professionals should consider asking school-aged children and adolescents with type 1 diabetes, **B** type 2 diabetes, **E** and other forms of diabetes, **E** along with their parents or caregivers, about social adjustment (peer relationships) and school performance to determine whether further intervention is needed.

14.14 Offer adolescents with diabetes time by themselves with their health care professional(s) at a developmentally appropriate age. **E**

14.15 During adolescence, generally during the stage of pubertal growth and development, incorporate reproductive health and preconception counseling into routine diabetes clinic visits for all individuals of childbearing potential because of the risk of adverse pregnancy outcomes in this population. **A**

Rapid and dynamic cognitive, developmental, and emotional changes occur during childhood, adolescence, and emerging adulthood. Diabetes management places substantial burdens on the child, adolescent, and family, necessitating ongoing assessment of psychosocial status and social determinants of health during routine diabetes visits (41,42). Consider the impact of diabetes on quality of life as well as the development of behavioral health problems related to diabetes distress, symptoms of depression, symptoms of anxiety, fear of hypoglycemia and hyperglycemia, disordered eating behaviors, and eating disorders (41,43). Further, consideration of the sociocultural context and efforts to personalize diabetes management are of critical importance to minimize barriers to care, enhance participation, and maximize response to treatment. Screening for food insecurity, housing instability, and other barriers related to the social determinants of health should be part of routine pediatric diabetes care (44).

Children and adolescents with type 1 diabetes, type 2 diabetes, and other forms of diabetes experience high rates of diabetes distress, depression, anxiety disorders, disordered eating behaviors, and eating disorders (45–50). For this reason, routine psychosocial screening is recommended for children and adolescents with type 1, type 2, and other forms of diabetes. Diabetes distress is the emotional response to living with and managing a demanding chronic condition. Approximately 30% of children and adolescents with type 1 diabetes (45) and 24% of children and adolescents with type 2 diabetes (51) report clinically significant levels of diabetes distress. Screening for diabetes distress may begin as early as 7 or 8 years of age (43), using validated tools for

children, adolescents, and their parents or caregivers (52). The pooled prevalence of depression is 22.2% in children and adolescents with type 1 and type 2 diabetes (53). This high rate highlights the importance of routine depression screening in this population, beginning at age 12 years and continuing thereafter (54), particularly when treatment goals are not being met or when there are significant changes in medical status or life circumstances. The pooled prevalence of anxiety is 17.7% in children and adolescents with type 1 and type 2 diabetes (53), supporting routine screening for anxiety in children and adolescents aged 8 years and above (54). For children and adolescents experiencing severe and/or frequent hypoglycemia as well as substantial hyperglycemia, it is important to screen both their parents or caregivers and the individual with diabetes for fear of hypoglycemia (55). Children as young as 6 years old can provide reliable self-reports for fear of hypoglycemia (55). Lastly, screening for disordered eating behaviors is recommended when signs and symptoms (e.g., unexplained weight loss, hyperglycemia, and DKA) and/or behavioral and emotional indicators (e.g., secrecy around eating and excessive concern about weight) are present using validated screening tools (56). Children and adolescents with type 1 diabetes are at increased risk for disordered eating behaviors and clinical eating disorders, which can lead to serious short- and long-term complications affecting diabetes outcomes and overall health. A particularly dangerous behavior in this population is insulin omission for weight loss (57). In the SEARCH study, disordered eating behaviors and insulin omission were also common among adolescents and young adults with type 2 diabetes, with 50.3% reporting disordered eating behaviors and 23% reporting insulin omission (58).

Given the complexity of psychosocial concerns in the management of diabetes in children and adolescents, collaboration between the diabetes health care team and a behavioral health professional is recommended. Early detection of diabetes distress, depression, anxiety, fear of hypoglycemia, and disordered eating behaviors can facilitate effective treatment options and help minimize adverse effects on diabetes management and health outcomes (41,43). When psychological symptoms are identified, referral to a behavioral health professional, ideally with experience in

pediatric diabetes, may be warranted. Such professionals can provide individualized, evidence-based behavioral health care services, including cognitive-behavioral therapy, mindfulness-based approaches, and other interventions, to improve psychosocial functioning in children and adolescents with diabetes (59–61).

The complexities of diabetes management require ongoing caregiver involvement in care throughout childhood and adolescence. Developmentally appropriate, supportive family teamwork between the growing child or adolescent and their parent(s) can help increase parent involvement in diabetes management and maintain or improve A1C in children and adolescents with A1C $\geq 8.0\%$ (62). It is appropriate to inquire about diabetes-specific family relationships, including family teamwork and conflict, during visits; health care professionals can both help families negotiate a plan and refer to an appropriate behavioral health professional for more in-depth support (63). Such professionals can conduct further assessments and deliver evidence-based behavioral interventions to support developmentally appropriate, collaborative family involvement in diabetes self-management (64,65). Monitoring of social adjustment (peer relationships) and school performance can facilitate both well-being and academic achievement (66,67). Diabetes management and glycemic levels may be associated with and impact academic progress and students' functioning in the school setting, which highlights the need for appropriate accommodations and access to diabetes-related support in school (68).

Decision-making involvement with children and adolescents regarding the adoption of management plan components can improve participation in diabetes self-management, technology adoption, and glycemic outcomes (69,70). Use of well-designed decision aids can engage children and adolescents in comprehensive, unbiased conversations with their diabetes care team about treatment options (71). Creating self-care contracts (72) and integrating technology that shares glucose data with the care team may facilitate shared decision-making and enhance care (73). Importantly, health care professionals working with children and adolescents who are not yet able to provide legal consent must balance clinical oversight with promoting developmentally appropriate independence. Recommendations include

providing education tailored to the developmental stage, encouraging gradual transfer of responsibility with self-care, guiding parental involvement as responsibilities change, teaching self-advocacy to prepare for transitions in care, and incorporating psychosocial support at all stages (67,74). Although cognitive abilities vary, the ethical position often adopted is the "mature minor rule," whereby children after age 12 or 13 years who can make an informed decision have the right to consent or withhold consent to general medical treatment, except in cases in which refusal would significantly endanger health (75). Importantly, not all children aged 12 or 13 years will be cognitively or emotionally prepared for transfer of daily self-care responsibilities. In these cases, emphasize assent to help ensure they are engaged in decision-making and to encourage them to take an active role in their self-management alongside caregivers.

Beginning at the onset of puberty or at diagnosis of diabetes for pubertal and postpubertal children and adolescents, all individuals with childbearing potential should receive education about the effective use of contraception to prevent unplanned pregnancy, as risks of fetal malformations are associated with elevated A1C. The TODAY study documented high rates of maternal complications during pregnancy and low rates of preconception counseling and contraception use in adolescents and young adults with type 2 diabetes (76). Preconception counseling using developmentally appropriate educational and behavioral strategies enables individuals of childbearing potential to make well-informed decisions (77). Preconception counseling resources tailored for adolescents are available at no cost through the ADA (78). Refer to the ADA position statement "Psychosocial Care for People With Diabetes" for further details (43).

The presence of behavioral health professionals on pediatric interprofessional teams emphasizes the importance of psychosocial care. These psychosocial factors are significantly related to self-management difficulties, elevated A1C, reduced quality of life, and higher rates of acute and chronic diabetes complications. A proactive, team-based approach to psychosocial screening, referral, and intervention is essential to support the health of children and adolescents with diabetes.

Glycemic Monitoring, Insulin Delivery, and Goals

Recommendations

14.16 Continuous glucose monitoring (CGM) should be offered for diabetes management at diagnosis or as soon as possible in children and adolescents with diabetes who are capable of using the device safely (either by themselves or with caregivers). **A** The choice of device should be made based on the individual's and family's circumstances, desires, and needs.

14.17 Offer automated insulin delivery (AID) systems for diabetes management to children and adolescents with type 1 diabetes who are capable of using the device safely (either by themselves or with caregivers). Choice of device should be made based on the individual's and family's circumstances, desires, and needs. **A**

14.18 Offer open-loop insulin pump therapy for type 1 diabetes management to children and adolescents on multiple daily injections who are capable of using the device safely (either by themselves or with caregivers) if unable to use AID systems. Choice of device should be made based on the individual's and family's circumstances, desires, and needs. **A**

14.19 Students with diabetes must be supported at school in the use of diabetes technology, including CGM, insulin pumps, connected insulin pens, and AID systems, as prescribed by their diabetes care team. **C**

14.20 A1C goals must be individualized and reassessed over time. An A1C of $<7\%$ (<53 mmol/mol) is appropriate for most children and adolescents with diabetes. **B**

14.21 Less stringent A1C goals (such as $<7\%$ [<53 mmol/mol] or $<7.5\%$ [<58 mmol/mol]) may be appropriate for children and adolescents with diabetes who cannot articulate symptoms of hypoglycemia; have hypoglycemia unawareness; cannot access advanced insulin delivery technology and/or CGM; cannot check blood glucose regularly; or have nonglycemic factors that increase A1C. **B**

14.22 Even less stringent A1C goals may be appropriate for children and adolescents with diabetes and a history of severe hypoglycemia or limited life expectancy or where the harms of

treatment are greater than the benefits. **B**

14.23 Health care professionals may reasonably suggest more stringent A1C goals (such as <6.5% [<48 mmol/mol]) for selected children and adolescents with diabetes if they can be achieved without significant hypoglycemia, excessive weight gain, negative impacts on well-being or mental health, or undue burden of care or in those who have nonglycemic factors that decrease A1C (e.g., lower erythrocyte life span). Lower goals may also be appropriate during the honeymoon phase. **B**

14.24 For children and adolescents with diabetes, CGM metrics derived from CGM use over the most recent 14 days (or longer) are recommended to be used in conjunction with or without A1C whenever possible. **B**

Current standards for diabetes management reflect the need to minimize hyperglycemia as safely as possible. The Diabetes Control and Complications Trial (DCCT) (79), which did not enroll children <13 years of age, demonstrated that near normalization of blood glucose levels was more difficult to achieve in adolescents than in adults. Nevertheless, the increased use of continuous glucose monitoring (CGM) and AID systems, goal setting, and improved education have been associated with more children and adolescents reaching the blood glucose goals recommended by the ADA (80–82), particularly in families in which the parents or caregivers as well as the child or adolescent with diabetes participate jointly to perform the required diabetes-related tasks.

TYPE 1 DIABETES IN CHILDREN AND ADOLESCENTS

Lower A1C in adolescence and young adulthood is associated with a lower risk and rate of microvascular and macrovascular complications (83–85) and demonstrates the effects of metabolic memory (86–89). While an A1C goal of <7% (<53 mmol/mol) is appropriate for most children and adolescents with type 1 diabetes, as AID algorithms continue to improve and access to these systems expands, children and adolescents who can attain a goal of <6.5% (<48 mmol/mol) without adverse effects should be encouraged to do so. In

addition, achieving lower A1C levels is likely facilitated by setting lower A1C goals (80,90). Lower goals may be possible during the honeymoon phase of type 1 diabetes and in those with access to CGM and AID. The honeymoon phase, also referred to as partial remission, refers to a transient period of increased endogenous insulin secretion shortly after diagnosis when initiation of insulin therapy resolves glucotoxicity and allows for improved insulin secretion from residual β -cells. Recent data with newer devices indicate that the risk of hypoglycemia at lower A1C is less than it was before (81,82,91–94). However, special consideration should be given to the risk of hypoglycemia in young children (aged <6 years) who are often unable to recognize, articulate, and/or manage hypoglycemia. Registry data indicate that lower A1C goals can be achieved in children, including those aged <6 years, without increased risk of severe hypoglycemia (91). Please refer to section 6, “Glycemic Goals, Hypoglycemia, and Hyperglycemic Crisis,” for more information on glycemic assessment.

A strong relationship exists between the frequency of glucose monitoring and glycemic outcomes (95,96). Glucose levels for all children and adolescents with type 1 diabetes should be monitored multiple times daily using CGM whenever possible. For those who cannot access CGM and in case of inaccurate CGM readings, access to finger-stick blood glucose monitoring remains essential. Use of CGM soon after type 1 diabetes diagnosis is associated with improved A1C (97–99). Parents, caregivers, and children and adolescents should be offered initial and ongoing education and support for CGM use. Behavioral support may further improve ongoing CGM use (100). Metrics derived from CGM include percent time in range (TIR), time below target range (<70 mg/dL and <54 mg/dL), and time above target range (>180 mg/dL and >250 mg/dL) (101). While studies indicate a relationship between TIR and A1C (102,103), it is still uncertain what the ideal goal TIR should be for children, and further studies are needed. Though 14 days of CGM data has long been considered the gold standard, recent data indicate that 10 or more days of CGM data provides a reliable assessment of glycemia and that ≥ 7 days of data may be sufficient for children and adolescents attaining TIR goals (104,105). Please refer to section 7,

“Diabetes Technology,” for more information on the use of blood glucose meters, CGM, and insulin pumps. More information on insulin injection technique can be found in section 9, “Pharmacologic Approaches to Glycemic Treatment.”

In addition, type 1 diabetes and chronic hyperglycemia are associated with adverse effects on cognition during childhood and adolescence (7,106,107). Several factors, including young age, severe hypoglycemia at <6 years of age, DKA, and chronic hyperglycemia (106,108,109), may contribute to adverse effects on brain development and function. Attaining higher CGM TIR is associated with better structural brain development and better neurocognitive outcomes (108), further emphasizing the importance of achieving glycemic goals (7,108).

Key Concepts in Setting Glycemic Goals

- Glycemic goals should be individualized, and lower goals may be reasonable based on a benefit-risk assessment.
- Blood glucose goals should be modified in children with frequent hypoglycemia or hypoglycemia unawareness.

Exercise Needs Specific to Children and Adolescents With Type 1 Diabetes

Real-world data from the Type 1 Diabetes Exercise Initiative Pediatric (T1DEXIP) study reported that glucose levels tended to drop during exercise, most particularly in those with lower A1C, shorter diabetes duration, and less fear of hypoglycemia (110). Furthermore, glucose levels tended to be lowest at about 8–16 h postexercise in those with similar characteristics, and nocturnal hypoglycemia was more frequent when activity levels were higher and longer in duration (111). These data underscore the need for individualized management strategies for children and adolescents using insulin, including common sense precautions and understanding how to make insulin dose adjustments when undertaking exercise and physical activity. Accessible rapid-acting carbohydrates and frequent blood glucose monitoring before, during, and after exercise, with or without CGM maximize safety with exercise. As hyperglycemia can occur before, during, and after physical activity, it is important to ensure the elevated glucose level is not related to insulin deficiency, as that can lead to worsening hyperglycemia with exercise and DKA risk. Intense

activity should be postponed with marked hyperglycemia (glucose ≥ 350 mg/dL [≥ 19.4 mmol/L]), moderate to large urine ketones, and/or β -hydroxybutyrate > 1.5 mmol/L in the setting of insulin deficiency. Caution may be needed when β -hydroxybutyrate levels are ≥ 0.6 mmol/L in association with absence of insulin (112,113).

Prevention and treatment of hypoglycemia associated with physical activity includes decreasing prandial insulin for the meal or snack before and after exercise, and/or increasing carbohydrate intake. Children and adolescents using insulin pumps without automated insulin delivery (AID) can lower basal rates by ~ 10 – 50% or more or suspend for 1–2 h during exercise (111). Decreasing basal rates or long-acting insulin doses by $\sim 20\%$ after exercise may reduce delayed exercise-induced hypoglycemia (110). Using AID systems may improve TIR (70–180 mg/dL) during exercise, and children and adolescents can use device-specific settings that are more conservative or increase the glucose goal to prevent hypoglycemia with exercise (114).

Blood glucose goals prior to physical activity and exercise are 126–180 mg/dL (7.0–10.0 mmol/L) but should be individualized based on the insulin plan and type, intensity, and duration of activity (36,115). The accuracy of CGM systems varies depending on the type of exercise (116). Consider additional carbohydrate intake during and/or after exercise, depending on duration and intensity of physical activity, to prevent hypoglycemia. For low-to moderate-intensity aerobic activities (30–60 min), and if the child or adolescent is fasting, 10–15 g of carbohydrate may prevent hypoglycemia (36). The timing of carbohydrate intake before exercise is of great importance for children and adolescents using AID systems, as a rise in glucose may result in increased automated insulin delivery with a subsequent increase in the risk for hypoglycemia (113). After insulin boluses (relative hyperinsulinemia), consider 0.5–1.0 g of carbohydrates/kg per h of exercise (~ 30 – 60 g), similar to carbohydrate requirements for optimizing performance in athletes without type 1 diabetes (117,118).

For children and adolescents with type 1 diabetes and obesity, physical activity is a key component of diabetes care. Obesity is equally common in children and adolescents with or without type 1 diabetes.

Living with obesity is associated with a higher frequency of cardiovascular risk factors, and it disproportionately affects children and adolescents from racial and ethnic minoritized groups (e.g., Black and Latino children and adolescents) (9,119). Therefore, diabetes health care professionals should monitor weight status and encourage a healthy eating pattern, physical activity, and healthy weight as key components of pediatric type 1 diabetes care.

TYPE 2 DIABETES IN CHILDREN AND ADOLESCENTS

For information on risk-based screening for type 2 diabetes and prediabetes in children and adolescents, please refer to section 2, “Diagnosis and Classification of Diabetes.” For additional support for these recommendations, see the ADA position statement “Evaluation and Management of Youth-Onset Type 2 Diabetes” (3).

Screening and Diagnosis of Type 2 Diabetes in Children and Adolescents

Recommendations

14.25 Consider risk-based screening for prediabetes and/or type 2 diabetes after the onset of puberty or after 10 years of age, whichever occurs earlier, in children with overweight (BMI ≥ 85 th to < 95 th percentile) or obesity (BMI ≥ 95 th percentile) and who have one or more additional risk factor for diabetes (see **Table 2.5** for evidence grading of other risk factors).

14.26 If screening is normal, repeat screening at a minimum of 2-year intervals **E** or more frequently if BMI is increasing. **C**

14.27 Fasting plasma glucose, 2-h plasma glucose during a 75-g oral glucose tolerance test, elevated random glucose with symptoms of hyperglycemia, and/or A1C can be used to test for prediabetes or type 2 diabetes in children and adolescents. **B**

14.28 Children and adolescents with overweight or obesity with hyperglycemia in whom the diagnosis of type 2 diabetes is being considered should have a panel of pancreatic autoantibodies tested to exclude the possibility of autoimmune type 1 diabetes. **B**

In recent years, incidence and prevalence of type 2 diabetes in children and

adolescents have increased dramatically, especially in historically marginalized communities (120). A few studies suggest oral glucose tolerance tests or fasting plasma glucose values as more suitable diagnostic tests than A1C in the pediatric population, especially among certain ethnicities (121), while fasting glucose alone may overdiagnose diabetes in children and adolescents (122,123). In addition, many studies that compare A1C to fasting plasma glucose or oral glucose tolerance test do not consider that diabetes diagnostic criteria are based on long-term health outcomes, and validations are not currently available in the pediatric population (124). An analysis of National Health and Nutrition Examination Survey (NHANES) data suggests using A1C for screening of high-risk children and adolescents due to association with cardiometabolic risk (125). While the ADA acknowledges there are limitations in the data supporting A1C for diagnosing type 2 diabetes in children and adolescents, the ADA continues to recommend its use in this population (121). A1C is not recommended for diagnosis of diabetes in children and adolescents with cystic fibrosis or symptoms suggestive of acute onset of type 1 diabetes, and only A1C assays without interference are appropriate for children and adolescents with hemoglobinopathies.

Diagnostic Challenges: Overlap of Type 1 and Type 2 Diabetes

Given the current obesity epidemic, distinguishing between type 1 and type 2 diabetes in children can be difficult. Overweight and obesity are common in children with type 1 diabetes (126), and diabetes-associated autoantibodies and ketosis may be present in pediatric individuals with clinical features of type 2 diabetes (including obesity and acanthosis nigricans) (122). The presence of islet autoantibodies has been associated with faster progression to insulin deficiency (122). At the onset of diabetes, DKA occurs in $\sim 11\%$ of children and adolescents aged 10–19 years with type 2 diabetes (127). Although uncommon, type 2 diabetes has been observed in prepubertal children under the age of 10 years, thus it should be part of the differential in children with suggestive symptoms (128). Finally, obesity contributes to the development of type 1 diabetes in some individuals (129), which further blurs

the lines between diabetes types. We must acknowledge that people with type 1 diabetes can also experience weight gain and insulin resistance. However, accurate diagnosis is critical, as treatment plans, educational approaches, nutrition advice, and outcomes differ markedly between individuals with predominantly insulin resistance versus absolute insulinopenia phenotypes. It must also be acknowledged that overlap between type 1 and type 2 diabetes can exist, and both insulin resistance and absolute insulinopenia should be treated appropriately (130). The significant diagnostic difficulties posed by maturity-onset diabetes of the young are discussed in section 2, "Diagnosis and Classification of Diabetes." In addition, there are rare and atypical diabetes cases that represent a challenge for clinicians and researchers.

Management

Lifestyle Management

Recommendations

14.29 Provide children and adolescents with overweight or obesity and type 2 diabetes and their families with developmentally and culturally appropriate comprehensive lifestyle programs integrated with diabetes management to achieve at least a 7–10% decrease in excess weight. **B**

14.30 Given the necessity of long-term weight management for children and adolescents with diabetes, lifestyle intervention should be based on a chronic care model and offered in the context of diabetes care. **E**

Glycemic Goals

Recommendations

14.31 For children and adolescents with type 2 diabetes, glycemic status should be assessed at least every 3 months or as frequently as clinically indicated. **E**

14.32 Consider setting an A1C goal of <6.5% (<48 mmol/mol) for most children and adolescents with type 2 diabetes who have a low risk of hypoglycemia. For those at higher risk of hypoglycemia, A1C goals should be individualized as clinically appropriate. **C**

Exercise Needs Specific to Children and Adolescents With Type 2 Diabetes

Children and adolescents with type 2 diabetes and their families should receive education and support for physical activity, including achieving and maintaining a healthy weight (34). Family-based interventions have been reported to be effective on physical activity behavior (25,131). A family-centered approach to lifestyle modification, including those aspects focused on physical activity, is essential in children and adolescents with type 2 diabetes, and recommendations should be culturally appropriate and sensitive to family resources (see section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes").

Pharmacologic Management

Recommendations

14.33 Initiate pharmacologic therapy with consideration of multiple add-on therapies early on, in addition to behavioral counseling for healthful nutrition and physical activity changes, at diagnosis of type 2 diabetes. **A**

14.34 In individuals with incidentally diagnosed type 2 diabetes (A1C <8.5% [<69 mmol/mol] and asymptomatic), metformin is the initial pharmacologic treatment of choice unless contraindicated by kidney function. **A**

14.35 Children and adolescents with marked hyperglycemia (A1C $\geq 8.5\%$ [≥ 69 mmol/mol]) without acidosis at diagnosis should be treated initially with long-acting insulin while metformin is initiated and titrated. **B**

14.36 Initiate subcutaneous or intravenous insulin treatment in individuals with ketoacidosis to rapidly correct the hyperglycemia and the metabolic derangement. Once acidosis is resolved, metformin should be initiated while subcutaneous insulin therapy is continued. **A**

14.37 In individuals presenting with severe hyperglycemia (blood glucose ≥ 600 mg/dL [≥ 33.3 mmol/L]), consider assessment and treatment as needed for hyperglycemic hyperosmolar state. **A**

14.38 If individualized glycemic goals are not achieved or maintained with metformin (with or without long-acting insulin), glucagon-like peptide 1 receptor agonist (GLP-1 RA) and/or sodium–glucose cotransporter 2 inhibitor

(SGLT2i) should be considered in children and adolescents with type 2 diabetes of approved ages. **A**

14.39 For children and adolescents with type 2 diabetes not meeting individualized glycemic goals, consider maximizing noninsulin therapies (metformin, GLP-1 RA, and SGLT2i) before initiating and/or intensifying the insulin therapy plan. **E**

14.40 In individuals with type 2 diabetes initially treated with insulin and metformin and/or other glucose-lowering medications who are meeting glucose goals based on blood glucose monitoring or CGM, reducing or discontinuing insulin should be considered. **B**

Glycemic goals should be individualized, taking into consideration the long-term health benefits of more stringent goals and risk for adverse effects, such as hypoglycemia. A lower A1C goal of <6.5% in children and adolescents with type 2 diabetes compared with <7% recommended in type 1 diabetes is justified by a lower risk of hypoglycemia and higher risk of complications in children and adolescents with type 2 diabetes (13,132–136). Blood glucose monitoring plans should be individualized, taking into consideration the pharmacologic treatment of the person. Although data on CGM in children and adolescents with type 2 diabetes are sparse (137,138), CGM could be considered in individuals requiring frequent blood glucose monitoring for diabetes management as well as for those on insulin who are capable of using the device safely (either by themselves or with caregivers).

Current pharmacologic treatment options for children and adolescents with type 2 diabetes are limited to four approved drug classes: insulin, metformin, glucagon-like peptide 1 (GLP-1) receptor agonists, and sodium–glucose cotransporter 2 inhibitors (SGLT2is). Presentation with ketoacidosis or marked ketosis requires a period of insulin therapy until fasting and postprandial glycemia have been restored to normal or near-normal levels. Insulin pump therapy should be considered for those on long-term multiple daily injections who are able to safely manage the device. Initial treatment should also be with insulin when the distinction

between type 1 diabetes and type 2 diabetes is unclear and in individuals who have random blood glucose concentrations ≥ 250 mg/dL (≥ 13.9 mmol/L) and/or A1C $\geq 8.5\%$ (≥ 69 mmol/mol) (139). Metformin therapy should be added after resolution of ketosis or ketoacidosis.

When initial insulin treatment is not required, initiation of metformin is recommended as first-line therapy. The TODAY study found that metformin alone provided durable glycemic management (A1C $< 8\%$ [< 64 mmol/mol] for 6 months) only in approximately half of the participants (140). The Restoring Insulin Secretion (RISE) Consortium study did not demonstrate differences in measures of glucose or β -cell function preservation between metformin and insulin, but there was more weight gain with insulin (141).

To date, the TODAY study is the only trial combining lifestyle and metformin therapy in children and adolescents with type 2 diabetes; the combination did not perform better than metformin alone in achieving durable glycemic levels (139).

RCTs in children and adolescents have shown that GLP-1 receptor agonists are safe and effective for decreasing A1C (142–146) and promoting weight loss at higher doses approved for obesity (147). Use of GLP-1 receptor agonists can increase the frequency of gastrointestinal side effects and should not be used in individuals with a family history of medullary thyroid cancer. A recently published RCT of the dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 receptor agonist tirzepatide at doses of 5 mg and 10 mg weekly demonstrated clinically meaningful and statistically significant improvements in both A1C and weight at 30 weeks in the pooled tirzepatide group versus placebo with benefits sustained at 52 weeks (148). The most common adverse events were gastrointestinal, as seen in adult trials (148). There is no regulatory approval to date for pediatric use of tirzepatide.

In addition to GLP-1 receptor agonists, SGLT2is are well-studied drugs in adults with type 2 diabetes, and now empagliflozin, dapagliflozin, and canagliflozin are approved for use in children and adolescents aged 10–17 years with type 2 diabetes based on clinical trials showing clinically relevant decreases in A1C (149–152). These agents are well tolerated in children and adolescents with type 2 diabetes with a low risk of severe hypoglycemia and

DKA. Individuals should be counseled on the increased risk of genitourinary infections.

In a multicenter double-blind, placebo-controlled trial of children and adolescents randomized at baseline to 10 mg empagliflozin, 5 mg linagliptin, or placebo, there was a significant reduction in A1C in the empagliflozin pooled group at 6 months (empagliflozin dose increased to 25 mg in a random manner in those with A1C $\geq 7\%$ at 14 weeks) compared with the placebo group; the linagliptin group did not demonstrate A1C benefit versus placebo (149). In a multicenter double-blind, placebo-controlled trial of dapagliflozin (151), children and adolescents with type 2 diabetes were randomized at baseline to 5 mg dapagliflozin, 2.5 mg saxagliptin, or placebo (with doses of dapagliflozin and saxagliptin increased in a random manner at week 14, namely, 10 mg dapagliflozin or 5 mg saxagliptin in those with A1C $\geq 7\%$). After 6 months, the pooled dapagliflozin versus placebo group had a significant reduction in A1C with no significant difference between the saxagliptin and placebo groups. Similarly, in the third SGLT2i randomized controlled clinical trial (152), children and adolescents with type 2 diabetes were randomized at baseline to 100 mg canagliflozin or placebo (with those in the canagliflozin group not achieving A1C $< 7\%$ at week 12 further randomized to continue at 100 mg or increase to 300 mg). After 6 months, the pooled canagliflozin versus placebo group demonstrated a significant reduction in A1C. In all three clinical trials of the SGLT2is, safety outcomes were assessed over 12 months and demonstrated safety profiles in children and adolescents matching those found in studies of adults with type 2 diabetes (149,151,152).

Metabolic Surgery

Recommendations

14.41 Consider metabolic surgery for the treatment of adolescents with type 2 diabetes who have class 2 obesity or higher (BMI 35 to < 40 kg/m² or 120% to $< 140\%$ percentile for age and sex, whichever is lower) and who have elevated A1C and/or serious comorbidities despite lifestyle and pharmacologic intervention. **A**

14.42 Metabolic surgery for adolescents with type 2 diabetes should

be performed only by an experienced surgeon working as part of a well-organized and engaged interprofessional team, including a surgeon, endocrinologist, registered dietitian nutritionist, behavioral health specialist, and nurse. **A**

The results of weight loss and lifestyle interventions for obesity in children and adolescents have been modest, and treatment options as adjuncts to lifestyle therapy are limited, although these findings do not include a more recently available dual GIP and GLP-1 receptor agonist. Recent U.S. Food and Drug Administration–approved medications for individuals age 12 years and older include phentermine and topiramate extended-release capsules and GLP-1 receptor agonists (147,153–155), with no pediatric regulatory approval of the dual GIP and GLP-1 receptor agonist. Metabolic surgery is now increasingly performed in adolescents with obesity. Small retrospective analyses and a prospective multicenter, non-randomized study suggest that bariatric or metabolic surgery have benefits in adolescents with obesity and type 2 diabetes like those observed in adults. Early follow-up studies indicate that adolescents experience similar degrees of weight loss compared with adults and even higher rates of type 2 diabetes and hypertension remission (156). A secondary data analysis from the Teen-Longitudinal Assessment of Bariatric Surgery (Teen-LABS) and TODAY studies suggests surgical treatment of adolescents with severe obesity and type 2 diabetes is associated with improved glycemia compared with the agents used in the TODAY study (157); however, no randomized trials have compared the effectiveness and safety of surgery with those of current treatment options in adolescents and particularly with the vertical sleeve gastrectomy, which is the most widely performed metabolic surgery in adolescents (158). The guidelines used as an indication for metabolic surgery in adolescents generally include class 2 obesity or higher (BMI 35 to < 40 kg/m² or 120% to $< 140\%$ of the 95th percentile for age and sex, whichever is lower), with comorbidities or BMI 40 kg/m² or higher with or without comorbidities (159–163). A number of groups, including the Pediatric Bariatric Study Group and Teen-LABS study, have

demonstrated the effectiveness of metabolic surgery in adolescents (159–163). The 10-year follow up data from Teen-LABS also demonstrates long-term durability of weight loss and comorbidity remission in adolescents (164). Additional long-term data on the rates of complications, reoperations, nutritional deficiencies, and diabetes recurrence are still needed. Furthermore, newer obesity pharmacotherapy trials are being performed in children and adolescents with the anticipation that positive results for weight loss in association with safety, including the recently published trial of tirzepatide (148), may supplant the need for metabolic surgery.

Prevention and Management of Diabetes Complications Among All Children and Adolescents With Diabetes

Recommendations **14.43–14.79**, regarding type 1 and type 2 diabetes complications, are presented in **Table 14.1** to facilitate direct comparison.

Autoimmune Conditions

Because of the increased frequency of other autoimmune diseases in type 1 diabetes, screening for thyroid dysfunction and celiac disease should be considered (165–169). Periodic screening in asymptomatic individuals has been recommended, but the optimal frequency of screening is unclear.

Although much less common than thyroid dysfunction and celiac disease, other autoimmune conditions, such as Addison disease (primary adrenal insufficiency), autoimmune hepatitis, autoimmune gastritis, dermatomyositis, premature ovarian failure, and myasthenia gravis, occur more commonly in the population with type 1 diabetes than in the general pediatric population and should be assessed and monitored as clinically indicated. In addition, relatives of individuals with type 1 diabetes should be offered testing for islet autoantibodies through research studies (e.g., Type 1 Diabetes TrialNet) and national programs for early diagnosis of preclinical type 1 diabetes (stages 1 and 2).

Thyroid Disease

Autoimmune thyroid disease is the most common autoimmune disorder associated with diabetes, occurring in 17–30% of individuals with type 1 diabetes (166,

170,171). At the time of diagnosis, ~25% of children with type 1 diabetes have thyroid autoantibodies (172), the presence of which is predictive of thyroid dysfunction—most commonly hypothyroidism, although hyperthyroidism occurs in ~0.5% of people with type 1 diabetes (173,174). For thyroid autoantibodies, a study from Sweden indicated that antithyroid peroxidase antibodies were more predictive of autoimmune thyroid disease than antithyroglobulin antibodies in multivariate analysis (175). Thyroid function tests may be misleading (nonthyroidal illness) if performed at the time of diagnosis owing to the effect of previous hyperglycemia, ketosis or ketoacidosis, weight loss, etc. Therefore, if performed at diagnosis and slightly abnormal, thyroid function tests should be repeated soon after a period of metabolic stability and achievement of glycemic goals. Subclinical hypothyroidism may be associated with an increased risk of symptomatic hypoglycemia and dyslipidemia (176,177) and a reduced linear growth rate. Hyperthyroidism alters glucose metabolism and usually causes deterioration of glycemia.

Celiac Disease

Celiac disease is an immune-mediated disorder that occurs with increased frequency in people with type 1 diabetes (1.6–16.4% of individuals compared with 0.3–1% in the general population) (165,168,169, 178–181). Screening people with type 1 diabetes for celiac disease is further justified by its association with osteoporosis, vitamin D deficiency, iron deficiency, growth failure, erratic glucose levels due to altered food absorption, and potential increased risk of retinopathy and albuminuria (182–184).

Screening for celiac disease includes measuring serum levels of IgA and tissue transglutaminase (tTG) IgA antibodies, or, with IgA deficiency, screening can include measuring tTG IgG antibodies or deamidated gliadin peptide IgG antibodies. Because most cases of celiac disease are diagnosed within the first 5 years after the diagnosis of type 1 diabetes, screening should be considered soon after diagnosis and repeated at 2 and then 5 years (179) or if clinical symptoms arise, such as poor growth or increased hypoglycemia (182).

Although celiac disease can be diagnosed more than 10 years after diabetes

diagnosis, there are insufficient data after 5 years to determine the optimal screening frequency. Gastroenterology guidelines on screening for celiac disease in children (not specific to children with type 1 diabetes) suggest that biopsy may not be necessary in symptomatic children with high antibody titers (i.e., >10 times the upper limit of normal) provided that further testing is performed (verification of endomysial antibody positivity on a separate blood sample) (185). About 95% of people with type 1 diabetes (186) and almost 99% of people with celiac disease carry HLA-DQ2 and/or HLA-DQ8 (187), so genetic testing may not be needed. In symptomatic children with type 1 diabetes and confirmed celiac disease, gluten-free eating patterns reduce symptoms and rates of hypoglycemia (188). The challenging eating plan restrictions associated with having both type 1 diabetes and celiac disease place an added burden on individuals. Therefore, a biopsy to confirm the diagnosis of celiac disease is recommended when the diagnosis is in question, especially in asymptomatic children and adolescents, before establishing a diagnosis of celiac disease and endorsing significant eating plan changes.

Management of Cardiovascular Risk Factors

Hypertension

Blood pressure measurements should be performed using the appropriate size cuff and with arm at heart level with the child or adolescent seated and relaxed. Elevated blood pressure should be confirmed on at least three separate days, and ambulatory blood pressure monitoring should be considered. Evaluation of secondary causes of hypertension should proceed as clinically indicated (189,190). Treatment is generally initiated with an ACE inhibitor, but an angiotensin receptor blocker can be used if the ACE inhibitor is not tolerated (e.g., due to cough) (191).

Dyslipidemia

Population-based studies estimate that 14–45% of children and adolescents with type 1 diabetes have two or more atherosclerotic cardiovascular disease (CVD) risk factors (192–194), and the prevalence of CVD risk factors increases with age (194) and among racial and ethnic minoritized groups (119), with girls having a higher risk burden than boys (193).

Table 14.1—Recommendations for diabetes-associated conditions in children and adolescents

	Type 1 diabetes recommendations	Type 2 diabetes recommendations	Type 1 and 2 diabetes recommendations
Dyslipidemia	14.43 For children and adolescents with type 1 diabetes, lipid screening should be performed soon after diagnosis, preferably after glycemia has improved and age is ≥2 years. If initial LDL cholesterol is ≤100 mg/dL (≤2.6 mmol/L), subsequent testing should be performed at 9–11 years of age B and repeated every 3 years. E	14.44 In children and adolescents with type 2 diabetes, lipid screening should be performed soon after diagnosis, preferably after glycemia has improved and annually thereafter. B	14.45 The LDL cholesterol goal is <100 mg/dL (<2.6 mmol/L). E 14.46 In children and adolescents with diabetes, if lipids are abnormal, initial therapy should consist of optimizing glycemia and medical nutritional therapy to limit calories from fat to 25–30% and saturated fat to <7%, limit cholesterol to <200 mg/day, avoid <i>trans</i> fats, and aim for ~10% calories from monounsaturated fats for elevated LDL. For elevated triglycerides, MNT should also focus on decreasing carbohydrate intake and increasing foods rich in n-3 fatty acids in addition to the above changes. A 14.47 In children and adolescents with diabetes, if LDL cholesterol remains >130 mg/dL (>3.4 mmol/L) after 6 months of nutrition intervention, initiate therapy with a statin, with a goal of LDL <100 mg/dL (<2.6 mmol/L). Due to the potential teratogenic effects, individuals of childbearing potential should receive reproductive counseling, and statins should be avoided in individuals of childbearing potential who are not using reliable contraception. B 14.48 In children and adolescents with diabetes, if triglycerides are >400 mg/dL (>4.7 mmol/L) fasting or >1,000 mg/dL (>11.6 mmol/L) nonfasting, optimize glycemia and begin fibrate, with a goal of <400 mg/dL (<4.7 mmol/L) fasting to reduce risk for pancreatitis. C
Hypertension			14.49 In children and adolescents with diabetes, BP should be measured at every clinic visit. In children and adolescents with high BP (BP ≥90th percentile for age, sex, and height or, in adolescents aged ≥13 years, ≥120/80 mmHg) on three separate measurements, ambulatory BP monitoring should be strongly considered. B 14.50 Excess weight increases cardiovascular event rates among people with diabetes and should be addressed with MNT, intensive lifestyle interventions focusing on dyslipidemia, hypertension, hyperglycemia along with adjunct pharmacotherapy, and/or bariatric surgery as appropriate. C 14.51 In children and adolescents with diabetes, after excluding secondary hypertension, treatment of elevated BP (defined as 90th to <95th percentile for age, sex, and height or, in adolescents aged ≥13 years, 120–129/<80 mmHg) is lifestyle modification focused on healthy nutrition, physical activity, sleep, and, if appropriate, weight management. C 14.52 For children and adolescents with diabetes, after excluding other causes, in addition to lifestyle modification, ACE inhibitors or ARBs should be started for treatment of

Continued on p. S309

Table 14.1—Continued

	Type 1 diabetes recommendations	Type 2 diabetes recommendations	Type 1 and 2 diabetes recommendations
Nephropathy	14.54 For children and adolescents with type 1 diabetes, nephropathy screening should be obtained at puberty or at age ≥ 11 years, whichever is earlier, once the youth have had diabetes for 5 years and annually thereafter. B	14.55 In children and adolescents with type 2 diabetes, nephropathy screening should be performed at the time of diagnosis and annually thereafter. An elevated UACR (>30 mg/g creatinine) should be confirmed on two of three samples. B	confirmed hypertension (defined as BP consistently ≥ 95th percentile for age, sex, and height or, in adolescents aged ≥ 13 years, BP $\geq 130/80$ mmHg). Due to the potential teratogenic effects, individuals of childbearing potential should receive reproductive counseling, and ACE inhibitors and ARBs should be avoided in individuals of childbearing potential who are not using reliable contraception. B 14.53 For children and adolescents the goal of hypertension treatment is BP <90th percentile for age, sex, and height or, in adolescents aged ≥ 13 years, BP $<130/80$ mmHg. C 14.56 In children and adolescents with diabetes, nephropathy screening should be performed with a random spot urine sample. Consider obtaining a morning sample if there are effects of exercise or orthostatic changes. An elevated UACR (>30 mg/g creatinine) should be confirmed on two of three samples over a 6-month period. B 14.57 Determine eGFR at the time of diagnosis and annually thereafter. E 14.58 In nonpregnant children and adolescents with diabetes, either an ACE inhibitor or an ARB is recommended for those with moderately increased albuminuria (UACR 30–299 mg/g creatinine) B and is strongly recommended for those with severely increased albuminuria (UACR ≥ 300 mg/g creatinine) and/or eGFR <60 mL/min/1.73 m² to maximally tolerated dose to prevent the progression of kidney disease and reduce cardiovascular events. A If one class is not tolerated, the other should be substituted. B Due to the potential teratogenic effects, individuals of childbearing potential should receive reproductive counseling, and ACE inhibitors and ARBs should be avoided in individuals of childbearing potential who are not using reliable contraception. B 14.59 For children and adolescents with nephropathy, continue monitoring (every 3–6 months and/or as indicated by UACR and eGFR) to detect disease progression. E 14.60 Refer to nephrology in case of uncertainty of etiology, worsening UACR, or decrease in eGFR. E
Retinopathy	14.61 An initial dilated and comprehensive diabetes eye examination is recommended once youth have had type 1 diabetes for 3–5 years, provided they are aged ≥ 11 years or puberty has started, whichever is earlier. B	14.62 Retinopathy screening in children and adolescents with type 2 diabetes should be performed at or soon after diagnosis and annually thereafter. C	14.63 Optimizing glycemia is recommended to decrease the risk or slow the progression of retinopathy. B

Continued on p. S310

Table 14.1—Continued

	Type 1 diabetes recommendations	Type 2 diabetes recommendations	Type 1 and 2 diabetes recommendations
Neuropathy	14.64 For children and adolescents with type 1 diabetes, after the initial examination, repeat dilated and comprehensive eye examination or retinal photography is recommended every 2 years. Less frequent examinations, every 4 years, may be acceptable on the advice of an eye care professional and based on risk factor assessment, including a history of A1C <8% (<64 mmol/mol). B 14.67 For children and adolescents with type 1 diabetes, consider an annual comprehensive foot exam at the start of puberty or at age ≥11 years, whichever is earlier, once the youth have had type 1 diabetes for 5 years. The examination should include inspection, assessment of foot pulses, pinprick, 10-g monofilament sensation tests, testing of vibration sensation using a 128-Hz tuning fork, and ankle reflex tests. B	14.65 Less frequent examination (every 2 years) using dilated eye examination or retinal photography may be considered if achieving glycemic goals and a normal eye exam. C 14.68 Screen children and adolescents with type 2 diabetes for the presence of neuropathy by foot examination at diagnosis and annually. The examination could include inspection, assessment of foot pulses, pinprick, 10-g monofilament sensation tests, testing of vibration sensation using a 128-Hz tuning fork, and ankle reflex tests. C	14.66 Programs that use nondilated retinal photography (with remote reading or use of a validated assessment tool) can be appropriate screening strategies for diabetic retinopathy. B
MASLD	14.69 Evaluate children and adolescents with type 2 diabetes for MASLD (by measuring AST and ALT) at diagnosis and annually thereafter. B 14.70 Consider referral to gastroenterology for persistently elevated or worsening transaminases. B		
Obstructive sleep apnea			14.71 In children and adolescents with diabetes, screening for symptoms of sleep apnea should be done at least annually, and referral to a pediatric sleep specialist is recommended for evaluation and a polysomnogram, if indicated. Obstructive sleep apnea should be treated when documented. B
Polycystic ovary syndrome			14.72 Evaluate for polycystic ovary syndrome in adolescent girls with diabetes, including laboratory studies, when clinically indicated. B 14.73 Metformin, in addition to lifestyle modification, is likely to improve the menstrual cyclicity and hyperandrogenism in adolescent girls with diabetes. E
Cardiovascular disease			14.74 Excessive weight gain increases cardiovascular event rates among children and adolescents with diabetes and should be addressed with lifestyle and obesity pharmacotherapy as appropriate. C

Continued on p. S311

Table 14.1—Continued

	Type 1 diabetes recommendations	Type 2 diabetes recommendations	Type 1 and 2 diabetes recommendations
Autoimmune conditions	14.75 In children and adolescents with type 1 diabetes, assess for autoimmune conditions outside of critical illness if clinically indicated. B		
Thyroid disease	14.76 In children and adolescents with type 1 diabetes, measure thyroid-stimulating hormone concentrations at diagnosis when clinically stable or soon after optimizing glycemia. If normal, suggest rechecking every 1–2 years or sooner if the individual has positive thyroid antibodies or develops symptoms or signs suggestive of thyroid dysfunction, thyromegaly, an abnormal growth rate, or unexplained glycemic variability. B		
Celiac disease	14.77 Screen children and adolescents with type 1 diabetes outside of critical illness for celiac disease by measuring IgA tissue tTG antibodies, with documentation of acceptable serum IgA levels for the local assay, soon after the diagnosis of diabetes, or IgG tTG and deamidated gliadin antibodies if IgA is deficient. B 14.78 Repeat screening for celiac disease within 2 years of type 1 diabetes diagnosis and then again after 5 years and consider more frequent screening in children and adolescents who have symptoms or a first-degree relative with celiac disease. B 14.79 Prescribe a gluten-free eating pattern for children and adolescents with confirmed celiac disease to avoid nutritional complications. Refer children and adolescents and their caregivers to a registered dietitian nutritionist experienced in managing both diabetes and celiac disease. B		

ARB, angiotensin receptor blocker; BP, blood pressure; eGFR, estimated glomerular filtration rate; MASLD, metabolic dysfunction–associated steatotic liver disease; MNT, medical nutrition therapy; tTG, tissue transglutaminase; UACR, urinary albumin-to-creatinine ratio.

Pathophysiology. The atherosclerotic process begins in childhood, and although atherosclerotic CVD events are not expected to occur during childhood, observations using a variety of methodologies show that children and adolescents with type 1 diabetes may have subclinical CVD within the first decade of diagnosis (195–197). Studies of carotid intima media thickness have yielded inconsistent results (190,191).

Screening. Diabetes predisposes the individual to the development of accelerated arteriosclerosis. Lipid evaluation for these individuals contributes to risk assessment and identifies an important proportion of those with dyslipidemia. Therefore, initial screening should be done soon after diagnosis. If the initial screen is normal, subsequent screening may be done at 9–11 years of age, which is a stable time for lipid assessment in children (198). Children and adolescents with a primary lipid disorder (e.g., familial hyperlipidemia) should be referred to a lipid specialist. Non-HDL cholesterol level has been identified as a significant predictor of the presence of atherosclerosis—as powerful as any other lipoprotein cholesterol measure in children and adolescents. For people with diabetes, non-HDL cholesterol level seems to be more predictive of persistent dyslipidemia and, therefore, atherosclerosis and future events than total cholesterol, LDL cholesterol, or HDL cholesterol level alone. A major advantage (199) of non-HDL cholesterol is that it can be accurately calculated in a nonfasting state and therefore is practical to obtain in clinical practice as a screening test (200). Youth with type 1 diabetes have a high prevalence of lipid abnormalities (192,199). Even if normal, screening should be repeated within 3 years, as A1C and other cardiovascular risk factors can change dramatically during adolescence (201).

Treatment. Pediatric lipid guidelines provide some guidance relevant to children and adolescents with type 1 diabetes and secondary dyslipidemia (190,191,202); however, there are few studies on modifying lipid levels in children and adolescents with type 1 diabetes. A 6-month trial of nutritional counseling produced a significant improvement in lipid levels (203); likewise, a lifestyle intervention trial with 6 months of exercise in adolescents demonstrated improvement in lipid levels (204). Data from the SEARCH study show

that improved glucose over a 2-year period is associated with a more favorable lipid profile; however, improved glycemia alone will not normalize lipids in children and adolescents with type 1 diabetes and dyslipidemia (205,206).

Although intervention data are sparse, the American Heart Association categorizes children and adolescents with type 1 diabetes in the highest tier for cardiovascular risk and recommends both lifestyle and pharmacologic treatment for those with elevated LDL cholesterol levels (207,208). Initial therapy should include a nutrition plan that restricts saturated fat to 7% of total calories and dietary cholesterol to 200 mg/day (198). Data from randomized clinical trials in children as young as 7 months of age indicate that this nutrition plan is safe and does not interfere with normal growth and development.

Long-term safety and cardiovascular outcome efficacy of statin therapy have been established for children with familial hypercholesterolemia (209). At the time of this writing, rosuvastatin is indicated for children as young as 6 years old (210). Statins should be avoided in individuals of childbearing age who are not using reliable contraception (see section 15, “Management of Diabetes in Pregnancy,” for more information). The multicenter, randomized, placebo-controlled Adolescent Type 1 Diabetes Cardio-Renal Intervention Trial (AdDIT) provides safety data on pharmacologic treatment with an ACE inhibitor and statin in adolescents with type 1 diabetes (190).

Microvascular Complications

Nephropathy

Data from 7,549 individuals <20 years of age in the T1D Exchange clinic registry emphasize the importance of meeting glycemic and blood pressure goals, particularly as diabetes duration increases, to reduce the risk of chronic kidney disease. The data also underscore the importance of routine screening to ensure early diagnosis and timely treatment of albuminuria (211). An estimation of glomerular filtration rate (GFR), calculated with GFR-estimating equations using serum creatinine, height, age, and sex (212), should be considered at baseline and repeated as indicated based on clinical status, age, diabetes duration, and therapies. Improved methods are needed to screen for early GFR loss, since estimated GFR is inaccurate

at GFR >75 mL/min/1.73 m² (212,213). The AdDIT study in adolescents with type 1 diabetes demonstrated the safety of ACE inhibitor treatment, but the treatment did not change the albumin-to-creatinine ratio over the course of the study (190).

Retinopathy

Retinopathy (like albuminuria) most commonly occurs after the onset of puberty and after 5–10 years of diabetes duration (214). It is currently recognized that there is a low risk of development of vision-threatening retinal lesions prior to 12 years of age (215,216). A 2019 publication based on the follow-up of the DCCT adolescent cohort supports a lower frequency of eye examinations than previously recommended, particularly in adolescents with A1C closer to the goal range (217,218). Autonomous artificial intelligence screening for diabetic retinopathy has been shown to increase access to this routine health maintenance (219). Referrals should be made to eye care professionals with expertise in diabetic retinopathy and experience in counseling pediatric individuals and families on the importance of prevention, early detection, and intervention.

Neuropathy

Diabetic neuropathy rarely occurs in prepubertal children or after only 1–2 years of diabetes (214), although data suggest a prevalence of distal peripheral neuropathy of 7% among 1,734 adolescents with type 1 diabetes in the SEARCH for Diabetes in Youth study and association with the presence of CVD risk factors (220,221). A comprehensive foot exam, including inspection, palpation of dorsalis pedis and posterior tibial pulses, and determination of proprioception, vibration, and monofilament sensation, should be performed annually along with an assessment of symptoms of neuropathic pain (221). Foot inspection can be performed at each visit to educate youth regarding the importance of foot care and often detects concurrent, unreported medical issues like tinea pedis or paronychia that would require treatment (see section 12, “Retinopathy, Neuropathy, and Foot Care”).

Type 2 Diabetes Comorbidities

Comorbidities may already be present at the time of diagnosis of type 2 diabetes in children and adolescents (12,222).

Therefore, blood pressure measurement, a fasting lipid panel, assessment of random urine albumin-to-creatinine ratio, foot examination for neuropathy, and a dilated eye examination should be performed at diagnosis. Additional medical conditions that may need to be addressed include polycystic ovary disease and other comorbidities associated with pediatric obesity, such as sleep apnea, hepatic steatosis, orthopedic complications, and psychosocial concerns. The ADA position statement “Evaluation and Management of Youth-Onset Type 2 Diabetes” (3) provides guidance on the prevention, screening, and treatment of type 2 diabetes and its comorbidities in children and adolescents.

Type 2 diabetes in children and adolescents is associated with significant microvascular and macrovascular risk burden and a substantial increase in the risk of cardiovascular morbidity and mortality at an earlier age than in those diagnosed later in life (13,223). The higher complication risk in earlier-onset type 2 diabetes is likely related to prolonged lifetime exposure to hyperglycemia and other atherogenic risk factors, including insulin resistance, dyslipidemia, hypertension, and chronic inflammation. There is a low risk of hypoglycemia in children and adolescents with type 2 diabetes, even if they are being treated with insulin (224), and there are high rates of chronic complications (133–135,224). These diabetes comorbidities also appear to be higher than those found in children and adolescents with type 1 diabetes despite shorter diabetes duration and lower A1C (222). In addition, the progression of vascular abnormalities appears to be more pronounced in children and adolescents-onset type 2 diabetes than with type 1 diabetes of similar duration, including ischemic heart disease and stroke (223).

In adolescents with type 2 diabetes and polycystic ovary syndrome, oral contraceptives are appropriate agents.

SUBSTANCE USE AMONG ALL CHILDREN AND ADOLESCENTS WITH DIABETES

Tobacco, Electronic Cigarettes, Alcohol, and Cannabis

Recommendations

14.80 Screen children and adolescents with diabetes for tobacco/nicotine vaping, substance use, and alcohol use

at diagnosis and regularly thereafter, **C** discourage use and provide appropriate referrals to cessation programs as needed. **A**

14.81 Advise all children, adolescents, and young adults with diabetes not to use cannabis recreationally in any form. **E**

The adverse health effects of smoking and use of tobacco products are well recognized with respect to future cancer and CVD risk. Despite this, rates of tobacco use are significantly higher among adolescents and young adults with type 1 and type 2 diabetes than those without diabetes (225,226). In children and adolescents with diabetes, it is important to avoid additional CVD risk factors. Smoking increases the risk of the onset of albuminuria; therefore, smoking avoidance is important to prevent both microvascular and macrovascular complications (199). Discouraging use of tobacco products, including vaping and electronic cigarettes (227,228), is an important part of routine diabetes care. Individuals with diabetes should be advised to avoid vaping and using electronic cigarettes, either as a way to stop smoking tobacco or as a recreational drug. In younger children, it is important to assess exposure to cigarette smoke in the home because of the adverse effects of secondhand smoke and to discourage them from ever smoking.

Alcohol use has implications for glycemic management and safety in adolescents and young adults with diabetes. Alcohol consumption can cause delayed hypoglycemia, typically 6 to 12 h after drinking (229). Further, alcohol increases the risk of hypoglycemia unawareness because symptoms of alcohol intoxication (e.g., slurred speech, drowsiness, confusion, difficulty walking) mimic symptoms of hypoglycemia (230). An untreated hypoglycemic event can result in loss of consciousness, seizure, and even death (230). For this reason, efforts are warranted to reduce alcohol use and increase education about the risks of alcohol use and strategies to minimize risks. A psychoeducational intervention for adolescents with chronic medical conditions, including type 1 diabetes, has demonstrated benefits for knowledge, understanding of perceived risks, and reduced use in some subgroups (231). See also section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes.”

Increased legalization and multiple formulations of cannabis products have increased the availability of these products. Among adolescents with type 1 diabetes, reported cannabis use ranged from 17.1% for past-month use to 23% for past 3-month use to 30.9% for lifetime use (231,232). Cannabis use increases the risk for hyperglycemic ketosis-cannabis hyperemesis syndrome in adults with type 1 diabetes (233,234), and emerging evidence in adolescents suggests a similar pattern (235). Hyperglycemic ketosis-cannabis hyperemesis syndrome is characterized by nausea that progresses to severe vomiting, leading to ketosis and alkalosis, and subsequently hyperglycemia (236). For adolescents with type 1 diabetes presenting with hyperglycemic emergencies, health care professionals should consider hyperglycemic ketosis-cannabis hyperemesis syndrome in individuals with pH ≥ 7.4 and bicarbonate >15 mmol/L in the presence of ketosis (236). Routine diabetes care should discourage the use of recreational cannabis in all forms. See section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes,” for more information about smoking cessation, tobacco, electronic cigarettes, and cannabis use in people with diabetes.

TRANSITION FROM PEDIATRIC TO ADULT CARE FOR ALL CHILDREN AND ADOLESCENTS WITH DIABETES

Recommendations

14.82 Diabetes care teams should implement transition preparation programs beginning in early adolescence and, at the latest, at least 1 year before the anticipated transfer from pediatric to adult health care. **E**

14.83 Interprofessional adult and pediatric health care teams should provide support and resources for adolescents, young adults, and their families prior to and during the transfer process from pediatric to adult health care. **C**

14.84 Pediatric diabetes specialists should partner with adolescents with diabetes and their caregivers to engage in shared decision-making for the timing of transfer to an adult diabetes specialist. There is no age-specific cutoff for young people with diabetes to transfer to an adult diabetes specialist. **E**

Care and close supervision of diabetes management are increasingly shifted from parents and other adults to the young people with type 1 or type 2 diabetes throughout childhood and adolescence. The shift from pediatric to adult health care professionals, however, often occurs abruptly as the older teen enters the next developmental stage, referred to as emerging adulthood (237), which is a critical period for young people who have diabetes. During this period of major life transitions, adolescents and young adults may begin to move out of their parents' or caregivers' homes and become increasingly responsible for their diabetes care. Their new responsibilities include self-management of their diabetes, scheduling and attending medical appointments, ensuring timely prescription refills, and financing health care once they are no longer covered by their parents' health insurance plans (ongoing coverage until age 26 years is currently available under provisions of the U.S. Affordable Care Act). In addition to lapses in health care, this is also a period associated with deterioration in glycemic stability; increased occurrence of acute complications; psychosocial, emotional, and behavioral challenges; and the emergence of chronic complications (238,239). The transfer period from pediatric to adult care is prone to fragmentation in health care delivery, which may adversely impact health care quality, cost, and outcomes (240). Worsening diabetes health outcomes during the transition to adult care and early adulthood have been documented (241,242).

Comprehensive and coordinated planning that begins in early adolescence is necessary to facilitate a seamless transition from pediatric to adult health care (238). Research on effective interventions to promote successful transition to adult care is limited, although there are promising developments that may improve attendance at follow-up appointments and lower hospitalizations (243,244). Use of transition coordinators, technology to support communication with young adults, and other interventions may be useful in addressing the identified needs and preferences of young adults for transition (245) and in supporting successful establishment in adult care settings (246–249). Given the behavioral, psychosocial, and developmental factors that relate to this transition, diabetes care teams addressing

transition should include clinicians certified diabetes care and education specialists, nurses, behavioral health professionals, registered dietitian nutritionists, and social workers (64,250). Resources to enhance social and peer support during the transition process may also be valuable (251). A comprehensive discussion regarding the challenges faced during this period, including specific recommendations, is found in the ADA position statement "Diabetes Care for Emerging Adults: Recommendations for Transition From Pediatric to Adult Diabetes Care Systems" (238). Ultimately, there is no age cutoff for adolescents and young adults with diabetes to transfer to adult diabetes care. The decision to transfer should be a collaborative process in which the young person with diabetes, their caregivers, and pediatric diabetes specialists discuss readiness, preferences, and concerns to ensure that the transfer aligns with the young person's needs and circumstances (244).

The Endocrine Society, in collaboration with the ADA and other organizations, has developed transition tools for clinicians and young people with diabetes and their families (238).

References

1. American Academy of Pediatrics. Recommended Child and Adolescent Immunization Schedule for Ages 18 Years or Younger. 2025. Accessed 21 August 2025. Available from https://aap2.silverchair-cdn.com/aap2/content_public/cms/resources/15585/0-18yrs-child-combined-schedule.pdf?Expires=2147483647&Signature=GDx7iGeLG8m9wTAJuJPB0P7WFuJPrQqPr9aQC4T1PzpYHKdkukTOTSPo3PSdsyNNum1CFFzShl7muP1SFgGgziJnYf7zAGKZZx4Ky4Q0BAhbsOXLby4c
2. Chiang JL, Maahs DM, Garvey KC, et al. Type 1 diabetes in children and adolescents: a position statement by the American Diabetes Association. *Diabetes Care* 2018;41:2026–2044
3. Arslanian S, Bacha F, Grey M, Marcus MD, White NH, Zeitler P. Evaluation and management of youth-onset type 2 diabetes: a position statement by the American Diabetes Association. *Diabetes Care* 2018;41:2648–2668
4. Lawrence JM, Divers J, Isom S, et al.; SEARCH for Diabetes in Youth Study Group. Trends in prevalence of type 1 and type 2 diabetes in children and adolescents in the US, 2001–2017. *JAMA* 2021;326:717–727
5. Leslie RD, Evans-Molina C, Freund-Brown J, et al. Adult-onset type 1 diabetes: current understanding and challenges. *Diabetes Care* 2021;44:2449–2456
6. Fang M, Wang D, Selvin E. Prevalence of type 1 diabetes among us children and adults by age, sex, race, and ethnicity. *JAMA* 2024;331:1411–1413
7. Ghetty S, Kuppermann N, Rewers A, et al.; Pediatric Emergency Care Applied Research Network (PECARN) DKA FLUID Study Group.

- Cognitive function following diabetic ketoacidosis in children with new-onset or previously diagnosed type 1 diabetes. *Diabetes Care* 2020;43:2768–2775
8. Markowitz JT, Garvey KC, Laffel LMB. Developmental changes in the roles of patients and families in type 1 diabetes management. *Curr Diabetes Rev* 2015;11:231–238
 9. Liu LL, Lawrence JM, Davis C, et al.; SEARCH for Diabetes in Youth Study Group. Prevalence of overweight and obesity in youth with diabetes in USA: the SEARCH for Diabetes in Youth study. *Pediatr Diabetes* 2010;11:4–11
 10. Imperatore G, Boyle JP, Thompson TJ, et al.; SEARCH for Diabetes in Youth Study Group. Projections of type 1 and type 2 diabetes burden in the U.S. population aged <20 years through 2050. *Diabetes Care* 2012;35:2515–2520
 11. Pettitt DJ, Talton J, Dabelea D, et al.; SEARCH for Diabetes in Youth Study Group. Prevalence of diabetes in U.S. youth in 2009: the SEARCH for diabetes in youth study. *Diabetes Care* 2014;37:402–408
 12. Copeland KC, Zeitler P, Geffner M, et al.; TODAY Study Group. Characteristics of adolescents and youth with recent-onset type 2 diabetes: the TODAY cohort at baseline. *J Clin Endocrinol Metab* 2011;96:159–167
 13. Bjornstad P, Drews KL, Caprio S, et al.; TODAY Study Group. Long-term complications in youth-onset type 2 diabetes. *N Engl J Med* 2021;385:416–426
 14. Naughton MJ, Ruggiero AM, Lawrence JM, et al.; SEARCH for Diabetes in Youth Study Group. Health-related quality of life of children and adolescents with type 1 or type 2 diabetes mellitus: SEARCH for Diabetes in Youth Study. *Arch Pediatr Adolesc Med* 2008;162:649–657
 15. Whalen DJ, Belden AC, Tillman R, Barch DM, Luby JL. Early adversity, psychopathology, and latent class profiles of global physical health from preschool through early adolescence. *Psychosom Med* 2016;78:1008–1018
 16. Cioana M, Deng J, Nadarajah A, et al. The prevalence of obesity among children with type 2 diabetes: a systematic review and meta-analysis. *JAMA Netw Open* 2022;5:e2247186
 17. Srinivasan S, Chen L, Todd J, et al. Erratum. the first genome-wide association study for type 2 diabetes in youth: the Progress in Diabetes Genetics in Youth (ProDiGY) consortium. *Diabetes* 2021;70:996–1005. *Diabetes* 2022;70:170
 18. Perng W, Oken E, Dabelea D. Developmental overnutrition and obesity and type 2 diabetes in offspring. *Diabetologia* 2019;62:1779–1788
 19. Cockcroft EJ, Clarke R, Dias RP, et al. Effectiveness of educational and psychoeducational self-management interventions in children and adolescents with type 1 diabetes: a systematic review and meta-analysis. *Pediatr Diabetes* 2024;2024:2921845
 20. Kurtzhals M, Bjerregaard AL, Hybschmann J, et al. A systematic review and meta-analysis of the child-level effects of family-based interventions for the prevention of type 2 diabetes mellitus. *Obes Rev* 2024;25:e13742
 21. Driscoll KA, Volkening LK, Haro H, et al. Are children with type 1 diabetes safe at school? Examining parent perceptions. *Pediatr Diabetes* 2015;16:613–620
 22. Kubota-Mishra E, Huang X, Minard CG, et al.; RADIANT Study Group. High prevalence of A-β+ ketosis-prone diabetes in children with type 2

diabetes and diabetic ketoacidosis at diagnosis: evidence from the Rare and Atypical Diabetes Network (RADIANT). *Pediatr Diabetes* 2024;2024:5907924

23. Muñoz-Pardeza J, López-Gil JF, Huerta-Urbe N, Hormazabal-Aguayo I, Izquierdo M, García-Hermoso A. Nonpharmacological interventions on glycated haemoglobin in youth with type 1 diabetes: a Bayesian network meta-analysis. *Cardiovasc Diabetol* 2024;23:230

24. U.S. Department of Agriculture and U.S. Department of Health and Human Services. *Dietary Guidelines for Americans, 2020–2025*. Accessed 30 August 2025. Available from https://www.dietaryguidelines.gov/sites/default/files/2021-03/Dietary_Guidelines_for_Americans-2020-2025.pdf

25. Carino M, New RH, Nguyen J, et al. Non-pharmacological management strategies for type 2 diabetes in children and young adults: a systematic review. *Diabetes Res Clin Pract* 2025;222:112045

26. Smith TA, Marlow AA, King BR, Smart CE. Insulin strategies for dietary fat and protein in type 1 diabetes: a systematic review. *Diabet Med* 2021;38:e14641

27. Paterson MA, King BR, Smart CEM, Smith T, Rafferty J, Lopez PE. Impact of dietary protein on postprandial glycaemic control and insulin requirements in Type 1 diabetes: a systematic review. *Diabet Med* 2019;36:1585–1599

28. Ringholm L, Nørgaard SK, Rytter A, Damm P, Mathiesen ER. Dietary advice to support glycaemic control and weight management in women with type 1 diabetes during pregnancy and breastfeeding. *Nutrients* 2022;14:4867

29. Steiman De Visser H, Fast I, Brunton N, et al. Cardiorespiratory fitness and physical activity in pediatric diabetes: a systemic review and meta-analysis. *JAMA Netw Open* 2024;7:e240235

30. García-Hermoso A, Ezzatvar Y, Huerta-Urbe N, et al. Effects of exercise training on glycaemic control in youths with type 1 diabetes: a systematic review and meta-analysis of randomised controlled trials. *Eur J Sport Sci* 2023;23:1056–1067

31. Huerta-Urbe N, Hormazabal-Aguayo IA, Izquierdo M, García-Hermoso A. Youth with type 1 diabetes mellitus are more inactive and sedentary than apparently healthy peers: a systematic review and meta-analysis. *Diabetes Res Clin Pract* 2023;200:110697

32. Migueles JH, Cadenas-Sanchez C, Lubans DR, et al. Effects of an exercise program on cardiometabolic and mental health in children with overweight or obesity: a secondary analysis of a randomized clinical trial. *JAMA Netw Open* 2023;6:e2324839

33. Slaght JL, Wicklow BA, Dart AB, et al. Physical activity and cardiometabolic health in adolescents with type 2 diabetes: a cross-sectional study. *BMJ Open Diabetes Res Care* 2021;9:e002134

34. U.S. Department of Health and Human Services. *Physical Activity Guidelines for Americans*, 2nd ed. Accessed 23 August 2025. Available from https://odphp.health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf

35. Colberg SR, Sigal RJ, Yardley JE, et al. Physical activity/exercise and diabetes: a position statement of the American Diabetes Association. *Diabetes Care* 2016;39:2065–2079

36. Moser O, Riddell MC, Eckstein ML, et al. Glucose management for exercise using continuous glucose monitoring (CGM) and intermittently

scanned CGM (isCGM) systems in type 1 diabetes: position statement of the European Association for the Study of Diabetes (EASD) and of the International Society for Pediatric and Adolescent Diabetes (ISPAD) endorsed by JDRF and supported by the American Diabetes Association (ADA). *Diabetologia* 2020;63:2501–2520

37. Adolfsson P, Taplin CE, Zaharieva DP, et al. ISPAD Clinical Practice Consensus Guidelines 2022: exercise in children and adolescents with diabetes. *Pediatr Diabetes* 2022;23:1341–1372

38. Cogen F, Rodriguez H, March CA, et al. Diabetes care in the school setting: a statement of the American Diabetes Association. *Diabetes Care* 2024;47:2050–2061

39. American Association of Diabetes Educators. Management of children with diabetes in the school setting. *Diabetes Educ* 2018;44:51–56

40. March C, Sherman J, Bannuru RR, et al. Care of young children with diabetes in the childcare and community setting: a statement of the American Diabetes Association. *Diabetes Care* 2023;46:2102–2111

41. Hilliard ME, De Wit M, Wasserman RM, et al. Screening and support for emotional burdens of youth with type 1 diabetes: strategies for diabetes care providers. *Pediatr Diabetes* 2018;19:534–543

42. Hill-Briggs F, Adler NE, Berkowitz SA, et al. Social determinants of health and diabetes: a scientific review. *Diabetes Care* 2020;44:258–279

43. Young-Hyman D, de Groot M, Hill-Briggs F, Gonzalez JS, Hood K, Peyrot M. Psychosocial care for people with diabetes: a position statement of the American Diabetes Association. *Diabetes Care* 2016;39:2126–2140

44. Odugbesan O, Wright T, Jones N-HY, et al. T1D Exchange Quality Improvement Collaborative. Increasing social determinants of health screening rates among six endocrinology centers across the United States: results from the T1D Exchange Quality Improvement Collaborative. *Clin Diabetes* 2024;42:49–55

45. Hagger V, Hendrieckx C, Sturt J, Skinner TC, Speight J. Diabetes distress among adolescents with type 1 diabetes: a systematic review. *Curr Diab Rep* 2016;16:9

46. Young V, Eiser C, Johnson B, et al. Eating problems in adolescents with type 1 diabetes: a systematic review with meta-analysis. *Diabet Med* 2013;30:189–198

47. Cooper MN, Lin A, Alvares GA, de Klerk NH, Jones TW, Davis EA. Psychiatric disorders during early adulthood in those with childhood onset type 1 diabetes: rates and clinical risk factors from population-based follow-up. *Pediatr Diabetes* 2017;18:599–606

48. Roberts AJ, Bao H, Qu P, et al. Mental health comorbidities in adolescents and young adults with type 2 diabetes. *J Pediatr Nurs* 2021;61:280–283

49. McVoy M, Hardin H, Fulchiero E, et al. Mental health comorbidity and youth onset type 2 diabetes: a systematic review of the literature. *Int J Psychiatry Med* 2023;58:37–55

50. Mateo K, Greenberg B, Valenzuela J. Disordered eating behaviors and eating disorders in youth with type 2 diabetes: a systematic review. *Diabetes Spectr* 2024;37:342–348

51. Trief PM, Uschner D, Tung M, et al. Diabetes distress in young adults with youth-onset type 2

diabetes: TODAY2 Study results. *Diabetes Care* 2022;45:529–537

52. Evans MA, Weil LEG, Shapiro JB, et al. Psychometric properties of the parent and child problem areas in diabetes measures. *J Pediatr Psychol* 2019;44:703–713

53. Akbarizadeh M, Naderi Far M, Ghaljaei F. Prevalence of depression and anxiety among children with type 1 and type 2 diabetes: a systematic review and meta-analysis. *World J Pediatr* 2022;18:16–26

54. Weitzman C, Guevara J, Curtin M, Macias M, Society for Developmental and Behavioral Pediatrics. Promoting optimal development: screening for mental health, emotional, and behavioral problems: clinical report. *Pediatrics* 2025;156:e2025073172

55. Andreopoulou O, Kostopoulou E, Kotanidou E, et al. Evaluation of the possible impact of the fear of hypoglycemia on diabetes management in children and adolescents with type 1 diabetes mellitus and their parents: a cross-sectional study. *Hormones (Athens)* 2024;23:419–428

56. Pursey KM, Hart M, Jenkins L, McEvoy M, Smart CE. Screening and identification of disordered eating in people with type 1 diabetes: a systematic review. *J Diabetes Complications* 2020;34:107522

57. Wisting L, Frøisland DH, Skriverhaug T, Dahl-Jørgensen K, Rø O. Disturbed eating behavior and omission of insulin in adolescents receiving intensified insulin treatment: a nationwide population-based study. *Diabetes Care* 2013;36:3382–3387

58. Nip ASY, Reboussin BA, Dabelea D, et al. SEARCH for Diabetes in Youth Study Group. Disordered eating behaviors in youth and young adults with type 1 or type 2 diabetes receiving insulin therapy: the SEARCH for Diabetes in Youth Study. *Diabetes Care* 2019;42:859–866

59. Inverso H, Moore HR, Lupini F, et al. Mindfulness-based interventions: focus on pediatric type 1 and type 2 diabetes. *Curr Diab Rep* 2022;22:493–500

60. Aljawarneh YM, Al-Qaissi NM, Ghunaim HY. Psychological interventions for adherence, metabolic control, and coping with stress in adolescents with type 1 diabetes: a systematic review. *World J Pediatr* 2020;16:456–470

61. Gulley LD, Shomaker LB, Kelly NR, et al. Examining cognitive-behavioral therapy change mechanisms for decreasing depression, weight, and insulin resistance in adolescent girls at risk for type 2 diabetes. *J Psychosom Res* 2022;157:110781

62. Katz ML, Volkeneing LK, Butler DA, Anderson BJ, Laffel LM. Family-based psychoeducation and Care Ambassador intervention to improve glycemic control in youth with type 1 diabetes: a randomized trial. *Pediatr Diabetes* 2014;15:142–150

63. Hickling A, Dingle GA, Barrett HL, Cobham VE. Systematic review: diabetes family conflict in young people with type 1 diabetes. *J Pediatr Psychol* 2021;46:1091–1109

64. Kichler JC, Harris MA, Weissberg-Benchell J. Contemporary roles of the pediatric psychologist in diabetes care. *Curr Diabetes Rev* 2015;11:210–221

65. Hilliard ME, Powell PW, Anderson BJ. Evidence-based behavioral interventions to promote diabetes management in children, adolescents, and families. *Am Psychol* 2016;71:590–601

66. Van Vleet M, Helgeson VS. Friend and peer relationships among youth with type 1 diabetes. In *Behavioral diabetes: social ecological perspectives for pediatric and adult populations*. Cham, Switzerland, Springer Nature Switzerland AG, 2020. pp. 121–138
67. Chiang JL, Kirkman MS, Laffel LMB, Peters AL; Type 1 Diabetes Sourcebook Authors. Type 1 diabetes through the life span: a position statement of the American Diabetes Association. *Diabetes Care* 2014;37:2034–2054
68. Knight MF, Perfect MM. Glycemic control influences on academic performance in youth with type 1 diabetes. *Sch Psychol* 2019;34:646–655
69. Miller VA, Jawad AF. Decision-making involvement and prediction of adherence in youth with type 1 diabetes: a cohort sequential study. *J Pediatr Psychol* 2019;44:61–71
70. Miller VA, Xiao R, Slick N, Feudtner C, Willi SM. Youth involvement in the decision to start CGM predicts subsequent CGM use. *Diabetes Care* 2020;43:2355–2361
71. Wysocki T, James L, Milkes A, et al. Electronically verified use of internet-based, multimedia decision aids by adolescents with type 1 diabetes and their caregivers. *MDM Policy Pract* 2018;3:2381468318769857
72. Hannon TS, Moore CM, Cheng ER, et al. Codesigned shared decision-making diabetes management plan tool for adolescents with type 1 diabetes mellitus and their parents: prototype development and pilot test. *J Particip Med* 2018;10:e8
73. Hannon TS, Yazel-Smith LG, Hatton AS, et al. Advancing diabetes management in adolescents: comparative effectiveness of mobile self-monitoring blood glucose technology and family-centered goal setting. *Pediatr Diabetes* 2018;19:776–781
74. Pugh A, Ritholz MD, Beverly EA. Similarities and Differences in Diabetes Diagnosis Stories Among Adults With Type 1 or Type 2 Diabetes in Appalachian Ohio. *Clin Diabetes* 2024;42:408–418
75. Weithorn LA. When does a minor's legal competence to make health care decisions matter? *Pediatrics* 2020;146:S25–S32
76. TODAY Study Group. Pregnancy outcomes in young women with youth-onset type 2 diabetes followed in the TODAY Study. *Diabetes Care* 2021;45:1038–1045
77. Peterson-Burch F, Abujaradeh H, Charache N, Fischl A, Charron-Prochownik D. Preconception counseling for adolescents and young adults with diabetes: a literature review of the past 10 years. *Curr Diab Rep* 2018;18:11
78. American Diabetes Association. Reproductive Health for Teen Girls with Diabetes. Accessed 31 August 2025. Available from <https://diabetes.org/health-wellness/sexual-health/reproductive-health-teen-girls-diabetes>
79. Nathan DM, Genuth S, Lachin J, et al.; Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *N Engl J Med* 1993;329:977–986
80. Gerhardsson P, Schwandt A, Witsch M, et al. The SWEET Project 10-year benchmarking in 19 countries worldwide is associated with improved HbA1c and increased use of diabetes technology in youth with type 1 diabetes. *Diabetes Technol Ther* 2021;23:491–499
81. Zimmermann AT, Lanzinger S, Kummernes SJ, et al. Treatment regimens and glycaemic outcomes in more than 100 000 children with type 1 diabetes (2013–22): a longitudinal analysis of data from paediatric diabetes registries. *Lancet Diabetes Endocrinol* 2025;13:47–56
82. Ebekozen O, Mungmode A, Sanchez J, et al. Longitudinal trends in glycemic outcomes and technology use for over 48,000 people with type 1 diabetes (2016–2022) from the T1D Exchange Quality Improvement Collaborative. *Diabetes Technol Ther* 2023;25:765–773
83. Diabetes Control and Complications Trial Research Group. Effect of intensive diabetes treatment on the development and progression of long-term complications in adolescents with insulin-dependent diabetes mellitus: Diabetes Control and Complications Trial. *J Pediatr* 1994;125:177–188
84. White NH, Cleary PA, Dahms W, Goldstein D, Malone J, Tamborlane WV; Diabetes Control and Complications Trial (DCCT)/Epidemiology of Diabetes Interventions and Complications (EDIC) Research Group. Beneficial effects of intensive therapy of diabetes during adolescence: outcomes after the conclusion of the Diabetes Control and Complications Trial (DCCT). *J Pediatr* 2001;139:804–812
85. Carlsen S, Skriverhaug T, Thue G, et al. Glycemic control and complications in patients with type 1 diabetes - a registry-based longitudinal study of adolescents and young adults. *Pediatr Diabetes* 2017;18:188–195
86. Genuth SM, Backlund J-YC, Bayless M, et al.; DCCT/EDIC Research Group. Effects of prior intensive versus conventional therapy and history of glycemia on cardiac function in type 1 diabetes in the DCCT/EDIC. *Diabetes* 2013;62:3561–3569
87. Writing Team for the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications Research Group. Sustained effect of intensive treatment of type 1 diabetes mellitus on development and progression of diabetic nephropathy: the Epidemiology of Diabetes Interventions and Complications (EDIC) study. *JAMA* 2003;290:2159–2167
88. Gubitosi-Klug RA, Sun W, Cleary PA, et al.; Writing Team for the DCCT/EDIC Research Group. Effects of prior intensive insulin therapy and risk factors on patient-reported visual function outcomes in the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications (DCCT/EDIC) cohort. *JAMA Ophthalmol* 2016;134:137–145
89. Orchard TJ, Nathan DM, Zinman B, et al.; Writing Group for the DCCT/EDIC Research Group. Association between 7 years of intensive treatment of type 1 diabetes and long-term mortality. *JAMA* 2015;313:45–53
90. Swift PGF, Skinner TC, de Beaufort CE, et al.; Hvidoere Study Group on Childhood Diabetes. Target setting in intensive insulin management is associated with metabolic control: the Hvidoere Childhood Diabetes Study Group Centre differences study 2005. *Pediatr Diabetes* 2010;11:271–278
91. Haynes A, Hermann JM, Miller KM, et al.; T1D Exchange, WACDD and DPV registries. Severe hypoglycemia rates are not associated with HbA1c: a cross-sectional analysis of 3 contemporary pediatric diabetes registry databases. *Pediatr Diabetes* 2017;18:643–650
92. Haynes A, Hermann JM, Clapin H, et al.; WACDD and DPV registries. Decreasing trends in mean HbA1c are not associated with increasing rates of severe hypoglycemia in children: a longitudinal analysis of two contemporary population-based pediatric type 1 diabetes registries from Australia and Germany/Austria between 1995 and 2016. *Diabetes Care* 2019;42:1630–1636
93. Karges B, Rosenbauer J, Kapellen T, et al. Hemoglobin A1c levels and risk of severe hypoglycemia in children and young adults with type 1 diabetes from Germany and Austria: a trend analysis in a cohort of 37,539 patients between 1995 and 2012. *PLoS Med* 2014;11:e1001742
94. Karges B, Rosenbauer J, Stahl-Peche A, et al. Hybrid closed-loop insulin therapy and risk of severe hypoglycaemia and diabetic ketoacidosis in young people (aged 2–20 years) with type 1 diabetes: a population-based study. *Lancet Diabetes Endocrinol* 2025;13:88–96
95. Miller KM, Beck RW, Bergenstal RM, et al.; T1D Exchange Clinic Network. Evidence of a strong association between frequency of self-monitoring of blood glucose and hemoglobin A1c levels in T1D exchange clinic registry participants. *Diabetes Care* 2013;36:2009–2014
96. Laffel LM, Kanapka LG, Beck RW, et al.; CDE10. Effect of continuous glucose monitoring on glycemic control in adolescents and young adults with type 1 diabetes: a randomized clinical trial. *JAMA* 2020;323:2388–2396
97. Prahalad P, Ding VY, Zaharieva DP, et al. Teamwork, targets, technology, and tight control in newly diagnosed type 1 diabetes: the Pilot 4T Study. *J Clin Endocrinol Metab* 2022;107:998–1008
98. Champakanath A, Akturk HK, Alonso GT, Snell-Bergeon JK, Shah VN. Continuous glucose monitoring initiation within first year of type 1 diabetes diagnosis is associated with improved glycemic outcomes: 7-year follow-up study. *Diabetes Care* 2022;45:750–753
99. Patton SR, Noser AE, Youngkin EM, Majidi S, Clements MA. Early initiation of diabetes devices relates to improved glycemic control in children with recent-onset type 1 diabetes mellitus. *Diabetes Technol Ther* 2019;21:379–384
100. Strategies to Enhance New CGM Use in Early Childhood (SENCE) Study Group. A randomized clinical trial assessing continuous glucose monitoring (CGM) use with standardized education with or without a family behavioral intervention compared with fingerstick blood glucose monitoring in very young children with type 1 diabetes. *Diabetes Care* 2021;44:464–472
101. Battelino T, Danne T, Bergenstal RM, et al. Clinical targets for continuous glucose monitoring data interpretation: recommendations from the International Consensus on Time in Range. *Diabetes Care* 2019;42:1593–1603
102. Vigersky RA, McMahon C. The relationship of hemoglobin A1C to time-in-range in patients with diabetes. *Diabetes Technol Ther* 2019;21:81–85
103. Petersson J, Åkesson K, Sundberg F, Särnblad S. Translating glycated hemoglobin A1c into time spent in glucose target range: a multicenter study. *Pediatr Diabetes* 2019;20:339–344
104. Riddlesworth TD, Beck RW, Gal RL, et al. Optimal sampling duration for continuous glucose

- monitoring to determine long-term glycemic control. *Diabetes Technol Ther* 2018;20:314–316
105. Gera S, Pearson A, Gallop RJ, Marks BE. Minimum continuous glucose monitor data required to assess glycemic control in youth with type 1 diabetes. *Diabetes Technol Ther* 2025;10.1089/dia.2025.0173
106. Mauras N, Buckingham B, White NH, et al.; Diabetes Research in Children Network (DirecNet). Impact of type 1 diabetes in the developing brain in children: a longitudinal study. *Diabetes Care* 2021;44:983–992
107. Foland-Ross LC, Tong G, Mauras N, et al.; Diabetes Research in Children Network (DirecNet). Brain function differences in children with type 1 diabetes: a functional MRI study of working memory. *Diabetes* 2020;69:1770–1778
108. Mauras N, Ma Q, Weinzimer SA, et al.; Diabetes Research in Children Network (DirecNet). Differences in white matter microstructure in children with type 1 diabetes persist during longitudinal follow-up: relation to dysglycemia. *Diabetes* 2025;74:1417–1426
109. Pourabbasi A, Tehrani-Doost M, Qavam SE, Arzaghi SM, Larijani B. Association of diabetes mellitus and structural changes in the central nervous system in children and adolescents: a systematic review. *J Diabetes Metab Disord* 2017;16:10
110. Riddell MC, Gal RL, Bergford S, et al. The acute effects of real-world physical activity on glycemia in adolescents with type 1 diabetes: the Type 1 Diabetes Exercise Initiative Pediatric (T1DEXIP) study. *Diabetes Care* 2024;47:132–139
111. Sherr JL, Bergford S, Gal RL, et al. Exploring factors that influence postexercise glycemia in youth with type 1 diabetes in the real world: the Type 1 Diabetes Exercise Initiative Pediatric (T1DEXIP) study. *Diabetes Care* 2024;47:849–857
112. Mehta SN, Volkening LK, Anderson BJ, et al.; Family Management of Childhood Diabetes Study Steering Committee. Dietary behaviors predict glycemic control in youth with type 1 diabetes. *Diabetes Care* 2008;31:1318–1320
113. Moser O, Zaharieva DP, Adolffson P, et al. The use of automated insulin delivery around physical activity and exercise in type 1 diabetes: a position statement of the European Association for the Study of Diabetes (EASD) and the International Society for Pediatric and Adolescent Diabetes (ISPAD). *Diabetologia* 2025;68:255–280
114. Eckstein ML, Weilguni B, Tauschmann M, et al. Time in range for closed-loop systems versus standard of care during physical exercise in people with type 1 diabetes: a systematic review and meta-analysis. *J Clin Med* 2021;10:2445
115. Riddell MC, Gallen IW, Smart CE, et al. Exercise management in type 1 diabetes: a consensus statement. *Lancet Diabetes Endocrinol* 2017;5:377–390
116. Da Prato G, Pasquini S, Rinaldi E, et al. Accuracy of CGM systems during continuous and interval exercise in adults with type 1 diabetes. *J Diabetes Sci Technol* 2022;16:1436–1443
117. Ajčević M, Candido R, Assaloni R, Accardo A, Francescato MP. Personalized approach for the management of exercise-related glycemic imbalances in type 1 diabetes: comparison with reference method. *J Diabetes Sci Technol* 2021;15:1153–1160
118. Baker LB, Rollo I, Stein KW, Jeukendrup AE. Acute effects of carbohydrate supplementation on intermittent sports performance. *Nutrients* 2015;7:5733–5763
119. Redondo MJ, Libman I, Cheng P, et al.; Pediatric Diabetes Consortium. Racial/ethnic minority youth with recent-onset type 1 diabetes have poor prognostic factors. *Diabetes Care* 2018;41:1017–1024
120. Tönnes T, Brinks R, Isom S, et al. Projections of type 1 and type 2 diabetes burden in the U.S. population aged <20 years through 2060: the SEARCH for Diabetes in Youth Study. *Diabetes Care* 2023;46:313–320
121. Buse JB, Kaufman FR, Linder B, Hirst K, El Ghormli L, Willi S, HEALTHY Study Group. Diabetes screening with hemoglobin A(1c) versus fasting plasma glucose in a multiethnic middle-school cohort. *Diabetes Care* 2013;36:429–435
122. Klingensmith GJ, Pyle L, Arslanian S, et al.; TODAY Study Group. The presence of GAD and IA-2 antibodies in youth with a type 2 diabetes phenotype: results from the TODAY study. *Diabetes Care* 2010;33:1970–1975
123. Hannon TS, Arslanian SA. The changing face of diabetes in youth: lessons learned from studies of type 2 diabetes. *Ann N Y Acad Sci* 2015;1353:113–137
124. Kapadia C, Zeitler P; Drugs and Therapeutics Committee of the Pediatric Endocrine Society. Hemoglobin A1c measurement for the diagnosis of type 2 diabetes in children. *Int J Pediatr Endocrinol* 2012;2012:31
125. Wallace AS, Wang D, Shin J-I, Selvin E. Screening and diagnosis of prediabetes and diabetes in US children and adolescents. *Pediatrics* 2020;146:e20200265
126. DuBose SN, Hermann JM, Tamborlane WV, et al.; Type 1 Diabetes Exchange Clinic Network and Diabetes Prospective Follow-up Registry. Obesity in youth with type 1 diabetes in Germany, Austria, and the United States. *J Pediatr* 2015;167:627–632 e621–624
127. Dabelea D, Rewers A, Stafford JM, et al.; SEARCH for Diabetes in Youth Study Group. Trends in the prevalence of ketoacidosis at diabetes diagnosis: the SEARCH for diabetes in youth study. *Pediatrics* 2014;133:e938–e945
128. Hutchins J, Barajas RA, Hale D, Escanave E, Lynch J. Type 2 diabetes in a 5-year-old and single center experience of type 2 diabetes in youth under 10. *Pediatr Diabetes* 2017;18:674–677
129. Ferrara CT, Geyer SM, Liu Y-F, et al.; Type 1 Diabetes TrialNet Study Group. Excess BMI in childhood: a modifiable risk factor for type 1 diabetes development? *Diabetes Care* 2017;40:698–701
130. Pozzilli P, Guglielmi C, Caprio S, Buzzetti R. Obesity, autoimmunity, and double diabetes in youth. *Diabetes Care* 2011;34 Suppl 2:S166–S170
131. Van Stappen V, Cardon G, De Craemer M, et al. The effect of a cluster-randomized controlled trial on lifestyle behaviors among families at risk for developing type 2 diabetes across Europe: the Feel4Diabetes-study. *Int J Behav Nutr Phys Act* 2021;18:86
132. TODAY Study Group. Safety and tolerability of the treatment of youth-onset type 2 diabetes: the TODAY experience. *Diabetes Care* 2013;36:1765–1771
133. TODAY Study Group. Retinopathy in youth with type 2 diabetes participating in the TODAY clinical trial. *Diabetes Care* 2013;36:1772–1774
134. TODAY Study Group. Lipid and inflammatory cardiovascular risk worsens over 3 years in youth with type 2 diabetes: the TODAY clinical trial. *Diabetes Care* 2013;36:1758–1764
135. TODAY Study Group. Rapid rise in hypertension and nephropathy in youth with type 2 diabetes: the TODAY clinical trial. *Diabetes Care* 2013;36:1735–1741
136. Zeitler P, Hirst K, Copeland KC, et al.; TODAY Study Group. HbA1c after a short period of monotherapy with metformin identifies durable glycemic control among adolescents with type 2 diabetes. *Diabetes Care* 2015;38:2285–2292
137. Chan CL. Use of continuous glucose monitoring in youth-onset type 2 diabetes. *Curr Diab Rep* 2017;17:66
138. Chesser H, Srinivasan S, Puckett C, Gitelman SE, Wong JC. Real-time continuous glucose monitoring in adolescents and young adults with type 2 diabetes can improve quality of life. *J Diabetes Sci Technol* 2024;18:911–919
139. Copeland KC, Silverstein J, Moore KR, et al.; American Academy of Pediatrics. Management of newly diagnosed type 2 diabetes mellitus (T2DM) in children and adolescents. *Pediatrics* 2013;131:364–382
140. Zeitler P, Hirst K, Pyle L, et al. A clinical trial to maintain glycemic control in youth with type 2 diabetes. *N Engl J Med* 2012;366:2247–2256
141. RISE Consortium. Impact of insulin and metformin versus metformin alone on β -cell function in youth with impaired glucose tolerance or recently diagnosed type 2 diabetes. *Diabetes Care* 2018;41:1717–1725
142. Tamborlane WV, Barrientos-Pérez M, Fainberg U, et al.; Ellipse Trial Investigators. Liraglutide in children and adolescents with type 2 diabetes. *N Engl J Med* 2019;381:637–646
143. U.S. Food and Drug Administration. FDA approves treatment for pediatric patients with type 2 diabetes - drug information update. 2021. Accessed 21 August 2025. Available from <https://content.govdelivery.com/accounts/USFDA/bulletins/2e98d66>
144. U.S. Food and Drug Administration. FDA approves new treatment for pediatric patients with type 2 diabetes. 2019. Accessed 21 August 2025. Available from <https://www.fda.gov/news-events/press-announcements/fda-approves-new-class-medicines-treat-pediatric-type-2-diabetes>
145. Tamborlane WV, Bishai R, Geller D, et al. Once-weekly exenatide in youth with type 2 diabetes. *Diabetes Care* 2022;45:1833–1840
146. Arslanian SA, Hannon T, Zeitler P, et al.; AWARD-PEDS Investigators. Once-weekly dulaglutide for the treatment of youths with type 2 diabetes. *N Engl J Med* 2022;387:433–443
147. Kelly AS, Auerbach P, Barrientos-Perez M, et al.; NN8022-4180 Trial Investigators. A randomized, controlled trial of liraglutide for adolescents with obesity. *N Engl J Med* 2020;382:2117–2128
148. Hannon TS, Chao LC, Barrientos-Pérez M, et al. Efficacy and safety of tirzepatide in children and adolescents with type 2 diabetes (SURPASS-PEDS): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2025;406:1484–1496
149. Laffel LM, Danne T, Klingensmith GJ, et al.; DINAMO Study Group. Efficacy and safety of the SGLT2 inhibitor empagliflozin versus placebo and the DPP-4 inhibitor linagliptin versus placebo in

- young people with type 2 diabetes (DINAMO): a multicentre, randomised, double-blind, parallel group, phase 3 trial. *Lancet Diabetes Endocrinol* 2023;11:169–181
150. Tamborlane WV, Laffel LM, Shehadeh N, et al. Efficacy and safety of dapagliflozin in children and young adults with type 2 diabetes: a prospective, multicentre, randomised, parallel group, phase 3 study. *Lancet Diabetes Endocrinol* 2022;10:341–350
151. Shehadeh N, Barrett T, Galassetti P, et al. Dapagliflozin or saxagliptin in pediatric type 2 diabetes. *NEJM Evid* 2023;2:EVID02300210
152. Nadgir U, Ali SR, Gogate J, Shaw W, Antunes J, Fonseca S. Treatment with canagliflozin versus placebo in children and adolescents with type 2 diabetes: a randomized clinical trial. *Ann Intern Med* 2025;178:1217–1226
153. Weghuber D, Kelly AS, Arslanian S. Once-weekly semaglutide in adolescents with obesity. reply. *N Engl J Med* 2023;388:1146
154. U.S. Food and Drug Administration. FDA approves weight management drug for patients aged 12 and older. 2021. Accessed 21 August 2025. Available from <https://www.fda.gov/drugs/drug-safety-and-availability/fda-approves-weight-management-drug-patients-aged-12-and-older>
155. U.S. Food and Drug Administration. FDA approves treatment for chronic weight management in pediatric patients aged 12 years and older. 2022. Accessed 27 August 2025. Available from <https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-treatment-chronic-weight-management-pediatric-patients-aged-12-years-and-older>
156. Beamish AJ, Ryan Harper E, Järholm K, Janson A, Olbers T. Long-term outcomes following adolescent metabolic and bariatric surgery. *J Clin Endocrinol Metab* 2023;108:2184–2192
157. Inge TH, Zeller M, Harmon C, et al. Teen-Longitudinal Assessment of Bariatric Surgery: methodological features of the first prospective multicenter study of adolescent bariatric surgery. *J Pediatr Surg* 2007;42:1969–1971
158. Rubino F, Nathan DM, Eckel RH, et al.; Delegates of the 2nd Diabetes Surgery Summit. Metabolic surgery in the treatment algorithm for type 2 diabetes: a joint statement by international diabetes organizations. *Diabetes Care* 2016;39:861–877
159. Torbahn G, Brauchmann J, Axon E, et al. Surgery for the treatment of obesity in children and adolescents. *Cochrane Database Syst Rev* 2022;9:CD011740
160. Michalsky MP, Inge TH, Simmons M, et al.; Teen-LABS Consortium. Cardiovascular risk factors in severely obese adolescents: the Teen Longitudinal Assessment of Bariatric Surgery (Teen-LABS) study. *JAMA Pediatr* 2015;169:438–444
161. Zeinoddini A, Heidari R, Talebpour M. Laparoscopic gastric plication in morbidly obese adolescents: a prospective study. *Surg Obes Relat Dis* 2014;10:1135–1139
162. Göthberg G, Gronowitz E, Flodmark C-E, et al. Laparoscopic Roux-en-Y gastric bypass in adolescents with morbid obesity—surgical aspects and clinical outcome. *Semin Pediatr Surg* 2014;23:11–16
163. Inge TH, Prigeon RL, Elder DA, et al. Insulin sensitivity and β -cell function improve after gastric bypass in severely obese adolescents. *J Pediatr* 2015;167:1042–1048.e1 e1041
164. Ryder JR, Jenkins TM, Xie C, et al. Ten-year outcomes after bariatric surgery in adolescents. *N Engl J Med* 2024;391:1656–1658
165. Warncke K, Fröhlich-Reiterer EE, Thon A, Hofer SE, Wiemann D, Holl RW; German BMBF Competence Network for Diabetes Mellitus. Polyendocrinopathy in children, adolescents, and young adults with type 1 diabetes: a multicenter analysis of 28,671 patients from the German/Austrian DPV-Wiss database. *Diabetes Care* 2010;33:2010–2012
166. Nederstigt C, Uitbeijse BS, Janssen LGM, Corssmit EPM, de Koning EJP, Dekkers OM. Associated auto-immune disease in type 1 diabetes patients: a systematic review and meta-analysis. *Eur J Endocrinol* 2019;180:135–144
167. Kozhakhmetova A, Wyatt RC, Caygill C, et al. A quarter of patients with type 1 diabetes have co-existing non-islet autoimmunity: the findings of a UK population-based family study. *Clin Exp Immunol* 2018;192:251–258
168. Hughes JW, Riddlesworth TD, DiMeglio LA, Miller KM, Rickels MR, McGill JB; T1D Exchange Clinic Network. Autoimmune diseases in children and adults with type 1 diabetes from the T1D Exchange Clinic Registry. *J Clin Endocrinol Metab* 2016;101:4931–4937
169. Kahaly GJ, Hansen MP. Type 1 diabetes associated autoimmunity. *Autoimmun Rev* 2016;15:644–648
170. Roldán MB, Alonso M, Barrio R. Thyroid autoimmunity in children and adolescents with type 1 diabetes mellitus. *Diabetes Nutr Metab* 1999;12:27–31
171. Shun CB, Donaghue KC, Phelan H, Twigg SM, Craig ME. Thyroid autoimmunity in type 1 diabetes: systematic review and meta-analysis. *Diabet Med* 2014;31:126–135
172. Triolo TM, Armstrong TK, McFann K, et al. Additional autoimmune disease found in 33% of patients at type 1 diabetes onset. *Diabetes Care* 2011;34:1211–1213
173. Kordonouri O, Deiss D, Danne T, Dorow A, Bassir C, Grüters-Kieslich A. Predictivity of thyroid autoantibodies for the development of thyroid disorders in children and adolescents with type 1 diabetes. *Diabet Med* 2002;19:518–521
174. Dost A, Rohrer TR, Fröhlich-Reiterer E, et al.; DPV Initiative and the German Competence Network Diabetes Mellitus. Hyperthyroidism in 276 children and adolescents with type 1 diabetes from Germany and Austria. *Horm Res Paediatr* 2015;84:190–198
175. Jonsdottir B, Larsson C, Carlsson A, et al.; Better Diabetes Diagnosis Study Group. Thyroid and islet autoantibodies predict autoimmune thyroid disease at type 1 diabetes diagnosis. *J Clin Endocrinol Metab* 2017;102:1277–1285
176. Mohn A, Di Michele S, Di Luzio R, Tumini S, Chiarelli F. The effect of subclinical hypothyroidism on metabolic control in children and adolescents with type 1 diabetes mellitus. *Diabet Med* 2002;19:70–73
177. Denzer C, Karges B, Näke A, et al.; DPV Initiative and the BMBF-Competence Network Diabetes Mellitus. Subclinical hypothyroidism and dyslipidemia in children and adolescents with type 1 diabetes mellitus. *Eur J Endocrinol* 2013;168:601–608
178. Holmes GKT. Screening for coeliac disease in type 1 diabetes. *Arch Dis Child* 2002;87:495–498
179. Pham-Short A, Donaghue KC, Ambler G, Phelan H, Twigg S, Craig ME. Screening for celiac disease in type 1 diabetes: a systematic review. *Pediatrics* 2015;136:e170–176.e176
180. Cerutti F, Bruno G, Chiarelli F, Lorini R, Meschi F, Sacchetti C; Diabetes Study Group of the Italian Society of Pediatric Endocrinology and Diabetology. Younger age at onset and sex predict celiac disease in children and adolescents with type 1 diabetes: an Italian multicenter study. *Diabetes Care* 2004;27:1294–1298
181. Taczanowska A, Schwandt A, Amed S, et al. Celiac disease in children with type 1 diabetes varies around the world: an international, cross-sectional study of 57375 patients from the SWEET registry. *J Diabetes* 2021;13:448–457
182. Simmons JH, Foster NC, Riddlesworth TD, et al.; T1D Exchange Clinic Network. Sex- and age-dependent effects of celiac disease on growth and weight gain in children with type 1 diabetes: analysis of the Type 1 Diabetes Exchange Clinic Registry. *Pediatr Diabetes* 2018;19:741–748
183. Margoni D, Chouliaras G, Ducas G, et al. Bone health in children with celiac disease assessed by dual x-ray absorptiometry: effect of gluten-free diet and predictive value of serum biochemical indices. *J Pediatr Gastroenterol Nutr* 2012;54:680–684
184. Mollazadegan K, Kugelberg M, Montgomery SM, Sanders DS, Ludvigsson J, Ludvigsson JF. A population-based study of the risk of diabetic retinopathy in patients with type 1 diabetes and celiac disease. *Diabetes Care* 2013;36:316–321
185. Rubio-Tapia A, Hill ID, Semrad C, et al. American College of Gastroenterology guidelines update: diagnosis and management of celiac disease. *Am J Gastroenterol* 2023;118:59–76
186. Steck AK, Rewers MJ. Genetics of type 1 diabetes. *Clin Chem* 2011;57:176–185
187. Sahin Y. Celiac disease in children: a review of the literature. *World J Clin Pediatr* 2021;10:53–71
188. Abid N, McGlone O, Cardwell C, McCallion W, Carson D. Clinical and metabolic effects of gluten free diet in children with type 1 diabetes and coeliac disease. *Pediatr Diabetes* 2011;12:322–325
189. Flynn JT, Kaelber DC, Baker-Smith CM, et al.; Subcommittee on Screening and Management of High Blood Pressure in Children. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. *Pediatrics* 2017;140:e20171904
190. Marcovecchio ML, Chiesa ST, Bond S, et al.; AdDIT Study Group. ACE inhibitors and statins in adolescents with type 1 diabetes. *N Engl J Med* 2017;377:1733–1745
191. de Ferranti SD, de Boer IH, Fonseca V, et al. Type 1 diabetes mellitus and cardiovascular disease: a scientific statement from the American Heart Association and American Diabetes Association. *Diabetes Care* 2014;37:2843–2863
192. Rodriguez BL, Fujimoto WY, Mayer-Davis EJ, et al. Prevalence of cardiovascular disease risk factors in U.S. children and adolescents with diabetes: the SEARCH for diabetes in youth study. *Diabetes Care* 2006;29:1891–1896
193. Margeisdottir HD, Larsen JR, Brunborg C, Overby NC, Dahl-Jørgensen K; Norwegian Study Group for Childhood Diabetes. High prevalence of cardiovascular risk factors in children and

- adolescents with type 1 diabetes: a population-based study. *Diabetologia* 2008;51:554–561
194. Schwab KO, Doerfer J, Hecker W, et al.; DPV Initiative of the German Working Group for Pediatric Diabetology. Spectrum and prevalence of atherogenic risk factors in 27,358 children, adolescents, and young adults with type 1 diabetes: cross-sectional data from the German diabetes documentation and quality management system (DPV). *Diabetes Care* 2006;29:218–225
 195. Singh TP, Groehn H, Kazmers A. Vascular function and carotid intimal-medial thickness in children with insulin-dependent diabetes mellitus. *J Am Coll Cardiol* 2003;41:661–665
 196. Haller MJ, Stein J, Shuster J, et al. Peripheral artery tonometry demonstrates altered endothelial function in children with type 1 diabetes. *Pediatr Diabetes* 2007;8:193–198
 197. Urbina EM, Wadwa RP, Davis C, et al. Prevalence of increased arterial stiffness in children with type 1 diabetes mellitus differs by measurement site and sex: the SEARCH for Diabetes in Youth Study. *J Pediatr* 2010;156:731–737, 737.e1 e731
 198. Expert Panel on Integrated Guidelines for Cardiovascular Health and Risk Reduction in Children and Adolescents; National Heart, Lung, and Blood Institute. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: summary report. *Pediatrics* 2011;128(Suppl 5):S213–S256
 199. Kershner AK, Daniels SR, Imperatore G, et al. Lipid abnormalities are prevalent in youth with type 1 and type 2 diabetes: the SEARCH for Diabetes in Youth Study. *J Pediatr* 2006;149:314–319
 200. Blaha MJ, Blumenthal RS, Brinton EA, Jacobson TA; National Lipid Association Taskforce on Non-HDL Cholesterol. The importance of non-HDL cholesterol reporting in lipid management. *J Clin Lipidol* 2008;2:267–273
 201. Maahs DM, Hermann JM, DuBose SN, et al.; T1D Exchange Clinic Network. Contrasting the clinical care and outcomes of 2,622 children with type 1 diabetes less than 6 years of age in the United States T1D Exchange and German/Austrian DPV registries. *Diabetologia* 2014;57:1578–1585
 202. Daniels SR, Greer FR; Committee on Nutrition. Lipid screening and cardiovascular health in childhood. *Pediatrics* 2008;122:198–208
 203. Cadario F, Prodam F, Pasqualicchio S, et al. Lipid profile and nutritional intake in children and adolescents with type 1 diabetes improve after a structured dietician training to a Mediterranean-style diet. *J Endocrinol Invest* 2012;35:160–168
 204. Salem MA, AboElAsrar MA, Elbarbary NS, Elhilaly RA, Refaat YM. Is exercise a therapeutic tool for improvement of cardiovascular risk factors in adolescents with type 1 diabetes mellitus? A randomised controlled trial. *Diabetol Metab Syndr* 2010;2:47
 205. Maahs DM, Dabelea D, D'Agostino RB, Jr, et al.; SEARCH for Diabetes in Youth Study. Glucose control predicts 2-year change in lipid profile in youth with type 1 diabetes. *J Pediatr* 2013;162:101–107.e1 e101
 206. Katz ML, Kollman CR, Dougher CE, Mubasher M, Laffel LMB. Influence of HbA1c and BMI on lipid trajectories in youths and young adults with type 1 diabetes. *Diabetes Care* 2017;40:30–37
 207. Kavey R-EW, Allada V, Daniels SR, et al.; Interdisciplinary Working Group on Quality of Care and Outcomes Research. Cardiovascular risk reduction in high-risk pediatric patients: a scientific statement from the American Heart Association Expert Panel on Population and Prevention Science; the Councils on Cardiovascular Disease in the Young, Epidemiology and Prevention, Nutrition, Physical Activity and Metabolism, High Blood Pressure Research, Cardiovascular Nursing, and the Kidney in Heart Disease; and the Interdisciplinary Working Group on Quality of Care and Outcomes Research: endorsed by the American Academy of Pediatrics. *Circulation* 2006;114:2710–2738
 208. McCrindle BW, Urbina EM, Dennison BA, et al.; American Heart Association Council on Cardiovascular Nursing. Drug therapy of high-risk lipid abnormalities in children and adolescents: a scientific statement from the American Heart Association Atherosclerosis, Hypertension, and Obesity in Youth Committee, Council of Cardiovascular Disease in the Young, with the Council on Cardiovascular Nursing. *Circulation* 2007;115:1948–1967
 209. Lurink IK, Wiegman A, Kusters DM, et al. 20-Year follow-up of statins in children with familial hypercholesterolemia. *N Engl J Med* 2019;381:1547–1556
 210. AstraZeneca Canada, Inc. Rosuvastatin product monograph, 2011. Accessed 31 August 2025. Available from <https://www.astrazeneca.ca/content/dam/az-ca/downloads/productinformation/crestor-product-monograph-en.pdf>
 211. Daniels M, DuBose SN, Maahs DM, et al.; T1D Exchange Clinic Network. Factors associated with microalbuminuria in 7,549 children and adolescents with type 1 diabetes in the T1D Exchange Clinic Registry. *Diabetes Care* 2013;36:2639–2645
 212. Schwartz GJ, Work DF. Measurement and estimation of GFR in children and adolescents. *Clin J Am Soc Nephrol* 2009;4:1832–1843
 213. Inker LA, Schmid CH, Tighiouart H, et al.; CKD-EPI Investigators. Estimating glomerular filtration rate from serum creatinine and cystatin C. *N Engl J Med* 2012;367:20–29
 214. Cho YH, Craig ME, Hing S, et al. Microvascular complications assessment in adolescents with 2- to 5-year duration of type 1 diabetes from 1990 to 2006. *Pediatr Diabetes* 2011;12:682–689
 215. Scanlon PH, Stratton IM, Bachmann MO, Jones C, Leese GP; Four Nations Diabetic Retinopathy Screening Study Group. Risk of diabetic retinopathy at first screen in children at 12 and 13 years of age. *Diabet Med* 2016;33:1655–1658
 216. Beauchamp G, Boyle CT, Tamborlane WV, et al.; T1D Exchange Clinic Network. Treatable diabetic retinopathy is extremely rare among pediatric T1D Exchange Clinic Registry participants. *Diabetes Care* 2016;39:e218–e219
 217. Nathan DM, Bebu I, Hainsworth D, et al.; DCCT/EDIC Research Group. Frequency of evidence-based screening for retinopathy in type 1 diabetes. *N Engl J Med* 2017;376:1507–1516
 218. Gubitosi-Klug RA, Bebu I, White NH, et al.; Diabetes Control and Complications Trial (DCCT)/Epidemiology of Diabetes Interventions and Complications (EDIC) Research Group. Screening eye exams in youth with type 1 diabetes under 18 years of age: once may be enough? *Pediatr Diabetes* 2019;20:743–749
 219. Wolf RM, Channa R, Liu TYA, et al. Autonomous artificial intelligence increases screening and follow-up for diabetic retinopathy in youth: the ACCESS randomized control trial. *Nat Commun* 2024;15:421
 220. Jaiswal M, Divers J, Dabelea D, et al. Prevalence of and risk factors for diabetic peripheral neuropathy in youth with type 1 and type 2 diabetes: SEARCH for Diabetes in Youth Study. *Diabetes Care* 2017;40:1226–1232
 221. Pop-Busui R, Boulton AJM, Feldman EL, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care* 2017;40:136–154
 222. Dabelea D, Stafford JM, Mayer-Davis EJ, et al.; SEARCH for Diabetes in Youth Research Group. Association of type 1 diabetes vs type 2 diabetes diagnosed during childhood and adolescence with complications during teenage years and young adulthood. *JAMA* 2017;317:825–835
 223. Song SH, Hardisty CA. Early onset type 2 diabetes mellitus: a harbinger for complications in later years—clinical observation from a secondary care cohort. *QJM* 2009;102:799–806
 224. Shah AS, Barrientos-Pérez M, Chang N, et al. ISPAD clinical practice consensus guidelines 2024: type 2 diabetes in children and adolescents. *Horm Res Paediatr* 2024;97:555–583
 225. Shah RD, Braffett BH, Tryggstad JB, et al.; TODAY Study Group. Cardiovascular risk factor progression in adolescents and young adults with youth-onset type 2 diabetes. *J Diabetes Complications* 2022;36:108123
 226. Creo A, Sriram S, Vaughan LE, Weaver AL, Lteif A, Kumar S. Risk of substance use disorders among adolescents and emerging adults with type 1 diabetes: a population-based cohort study. *Pediatr Diabetes* 2021;22:1143–1149
 227. Foxon F, Selya AS. Electronic cigarettes, nicotine use trends and use initiation ages among US adolescents from 1999 to 2018. *Addiction* 2020;115:2369–2378
 228. Veliz PT, McCabe SE, Evans-Polce RJ, Boyd CJ. Assessing how the history of e-cigarette and cigarette use are associated with the developmental course of marijuana use in a sample of United States adolescents. *Drug Alcohol Depend* 2020;216:108308
 229. Tetzschner R, Nørgaard K, Ranjan A. Effects of alcohol on plasma glucose and prevention of alcohol-induced hypoglycemia in type 1 diabetes—a systematic review with GRADE. *Diabetes Metab Res Rev* 2018;34:e2965
 230. Hölzen L, Schultes B, Meyhöfer SM, Meyhöfer S. Hypoglycemia unawareness—a review on pathophysiology and clinical implications. *Biomedicines* 2024;12:391
 231. Weitzman ER, Wisk LE, Minegishi M, et al. Effects of a patient-centered intervention to reduce alcohol use among youth with chronic medical conditions. *J Adolesc Health* 2022;71:S24–S33
 232. Hanna KM, Stupiansky NW, Weaver MT, Slaven JE, Stump TE. Alcohol use trajectories after high school graduation among emerging adults with type 1 diabetes. *J Adolesc Health* 2014;55:201–208
 233. Kinney GL, Akturk HK, Taylor DD, Foster NC, Shah VN. Cannabis use is associated with increased risk for diabetic ketoacidosis in adults with type 1

diabetes: findings from the T1D Exchange Clinic Registry. *Diabetes Care* 2020;43:247–249

234. Akturk HK, Taylor DD, Camsari UM, Rewers A, Kinney GL, Shah VN. Association between cannabis use and risk for diabetic ketoacidosis in adults with type 1 diabetes. *JAMA Intern Med* 2019;179:115–118

235. Pancer J, Dasgupta K. Effects of cannabis use in youth and young adults with type 1 diabetes: the highs, the lows, the don't knows. *Can J Diabetes* 2020;44:121–127

236. Akturk HK, Snell-Bergeon J, Kinney GL, Champakanath A, Monte A, Shah VN. Differentiating diabetic ketoacidosis and hyperglycemic ketosis due to cannabis hyperemesis syndrome in adults with type 1 diabetes. *Diabetes Care* 2022;45:481–483

237. Arnett JJ. Emerging adulthood. A theory of development from the late teens through the twenties. *Am Psychol* 2000;55:469–480

238. Peters A, Laffel L; American Diabetes Association Transitions Working Group. Diabetes care for emerging adults: recommendations for transition from pediatric to adult diabetes care systems: a position statement of the American Diabetes Association, with representation by the American College of Osteopathic Family Physicians, the American Academy of Pediatrics, the American Association of Clinical Endocrinologists, the American Osteopathic Association, the Centers for Disease Control and Prevention, Children with Diabetes, The Endocrine Society, the International Society for Pediatric and Adolescent Diabetes,

Juvenile Diabetes Research Foundation International, the National Diabetes Education Program, and the Pediatric Endocrine Society (formerly Lawson Wilkins Pediatric Endocrine Society). *Diabetes Care* 2011;34:2477–2485

239. Agarwal S, Raymond JK, Isom S, et al. Transfer from paediatric to adult care for young adults with type 2 diabetes: the SEARCH for Diabetes in Youth Study. *Diabet Med* 2018;35:504–512

240. Mays JA, Jackson KL, Derby TA, et al. An evaluation of recurrent diabetic ketoacidosis, fragmentation of care, and mortality across Chicago, Illinois. *Diabetes Care* 2016;39:1671–1676

241. Lotstein DS, Seid M, Klingensmith G, et al.; SEARCH for Diabetes in Youth Study Group. Transition from pediatric to adult care for youth diagnosed with type 1 diabetes in adolescence. *Pediatrics* 2013;131:e1062–e1070

242. Lyons SK, Becker DJ, Helgeson VS. Transfer from pediatric to adult health care: effects on diabetes outcomes. *Pediatr Diabetes* 2014;15:10–17

243. D'Amico RP, Pian TM, Buschur EO. Transition from pediatric to adult care for individuals with type 1 diabetes: opportunities and challenges. *Endocr Pract* 2023;29:279–285

244. Lal RA, Maahs DM, Dosiou C, Aye T, Basina M. The guided transfer of care improves adult clinic show rate. *Endocr Pract* 2020;26:508–513

245. Xie LF, Housni A, Nakhla M, et al. Adaptation of an adult web application for type 1 diabetes self-management to youth using the behavior change wheel to tailor the needs of

health care transition: qualitative interview study. *JMIR Diabetes* 2023;8:e42564

246. Butalia S, Crawford SG, McGuire KA, Dyjur DK, Mercer JR, Pacaud D. Improved transition to adult care in youth with type 1 diabetes: a pragmatic clinical trial. *Diabetologia* 2021;64:758–766

247. Spaic T, Robinson T, Goldbloom E, et al.; JDRF Canadian Clinical Trial CCTN1102 Study Group. Closing the gap: results of the multicenter canadian randomized controlled trial of structured transition in young adults with type 1 diabetes. *Diabetes Care* 2019;42:1018–1026

248. White M, O'Connell MA, Cameron FJ. Clinic attendance and disengagement of young adults with type 1 diabetes after transition of care from paediatric to adult services (TrACeD): a randomised, open-label, controlled trial. *Lancet Child Adolesc Health* 2017;1:274–283

249. Sequeira PA, Pyatak EA, Weigensberg MJ, et al. Let's Empower and Prepare (LEAP): evaluation of a structured transition program for young adults with type 1 diabetes. *Diabetes Care* 2015;38:1412–1419

250. Monaghan M, Baumann K. Type 1 diabetes: addressing the transition from pediatric to adult-oriented health care. *Res Rep Endocr Disord* 2016;6:31–40

251. Carreon SA, Duran B, Tang TS, et al. Here for you: a review of social support research in young adults with diabetes. *Diabetes Spectr* 2021;34:363–370